



School of Computing

Department of Computer Science and Engineering

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(WINTER SEMESTER)

LAB RECORD

10211CS224 - Full Stack Application Development

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YEAR/SEM : 3/6th

SLOT : E1

10211CS224 - Full Stack Application Development

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Task-1: Student Registration & Data Storage System

AIM: To develop a Student Registration System that allows users to enter, store, and manage student details efficiently using a web interface and a MySQL database.

ALGORITHM:

Step 1: Start

Step 2: Open the web application (index page)

Step 3: Display home page with option to go to registration page

Step 4: User navigates to the registration form

Step 5: User enters student details (like name, ID, course, etc.)

Step 6: Validate input fields

- Check if fields are empty
- Ensure correct data format

Step 7: Submit the form data to the server

Step 8: Server receives the data using Node.js

Step 9: Establish connection with MySQL database

- If connection fails → display error
- Else → proceed

Step 10: Insert student data into database table

Step 11: Confirm successful storage of data

Step 12: Display success message to user

Step 13: Allow user to view/manage stored data (if implemented)

Step 14: Stop

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>
```

```
<html>

<head>

<title>Student Portal</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="home-container">

<h1>Welcome to Student Portal</h1>

<p>Register and manage student information easily.</p>

<a href="registration.html" class="start-btn">Go to Registration</a>

</div>

</body>

</html>
```

REGISTRATION.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Student Registration</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

<h1>Student Registration</h1>

<button onclick="goHome()" class="home-btn">Home</button>

<form id="studentForm">

<label>Name</label>
```

```
<input type="text" id="name" placeholder="Enter your name" required>
<label>Email</label>
<input type="email" id="email" placeholder="Enter your email" required>
<label>Date of Birth</label>
<input type="date" id="dob" required>
<label>Department</label>
<select id="department" required>
<option value="">Select Department</option>
<option>CSE</option>
<option>IT</option>
<option>ECE</option>
<option>MECH</option>
</select>
<label>Phone</label>
<input type="text" id="phone" placeholder="Enter phone number" required>
<button type="submit">Register</button>
</form>
</div>
<script src="script.js"></script>
</body>
</html> STYLE.CSS: body{ font-family: 'Segoe UI', sans-serif;
background: linear-
gradient(135deg,#0f2027,#203a43,#2c5364); height:100vh;
display:flex; justify-
content:center; align-
```

```
items:center;
margin:0;
}
.container{ width:420px; padding:35px;
background:rgba(255,255,255,0.08);
border-radius:18px;    backdrop-filter:
blur(10px); box-shadow:0 10px 35px
rgba(0,0,0,0.6);
}
h1{ text-align:center;
color:white; margin-
bottom:25px;
}
h2{ text-
align:center;
color:white; margin-
top:30px;
}
form{ display:flex;
flex-direction:column;
}
label{ color:#ddd;
font-size:14px;
margin-bottom:5px;
margin-top:10px;
```

```
}  
  
input,select{  
width:100%;  
  
padding:12px;  
  
border:none; border-  
radius:8px;  
  
background:#27496d;  
  
color:white; font-  
size:14px;  
  
outline:none; margin-  
bottom:5px;  
  
}  
  
input:focus,select:focus{ box-  
shadow:0 0 8px #00eaff;  
  
background:#2c5d8a;  
  
}  
  
button{ margin-  
top:15px;  
  
padding:12px;  
  
border:none; border-  
radius:10px;  
  
background:linear-  
gradient(45deg,#ff4b  
2b,#ff416c);  
  
color:white; font-  
size:16px; font-
```

```
weight:bold;
cursor:pointer;
transition:0.3s;
}
button:hover{
transform:scale(1.05); box-
shadow:0 0 15px #ff416c;
}
#studentList{ margin-
top:15px; text-
align:center;
color:white; font-
size:15px;
}
.home-container{      text-align:center;
color:white;
background:rgba(255,255,255,0.08);
padding:50px;      border-radius:20px;
backdrop-filter:    blur(10px);    box-
shadow:0 10px 35px rgba(0,0,0,0.6);
width:420px;

}
.home-container h1{
font-size:32px;
margin-bottom:15px;
```



```

}

.home-container p{
margin-bottom:30px;
color:#ddd;
}

.start-btn{ text-decoration:none; padding:12px
25px; border-radius:10px; background:linear-
gradient(45deg,#ff4b2b,#ff416c); color:white; font-
weight:bold; transition:0.3s;
}

.start-btn:hover{
transform:scale(1.05); box-
shadow:0 0 15px #ff416c;
}

```

SCRIPT.JS:

```

const form = document.getElementById("studentForm") form.addEventListener("submit", async
(e) => {
e.preventDefault()    const    student    =    {    name:
document.getElementById("name").value,                email:
document.getElementById("email").value,                dob:
document.getElementById("dob").value,                department:
document.getElementById("department").value,    phone:
document.getElementById("phone").value
}

```

```

try{ const res = await
fetch("http://localhost:3000/addStudent",{
method:"POST", headers:{
"Content-Type":"application/json"
},
body:JSON.stringify(student)
})

const data = await
res.text() alert(data) //
clear form form.reset()

}catch(error){
console.log("Error:",error)
}
})

function goHome(){
window.location.href="index.html"
}

```

SERVER.JS:

```

const express =
require("express") const cors =
require("cors") const path =
require("path") const db =
require("./db") const app =
express() app.use(cors())
app.use(express.json())

```

```

// Serve static files (HTML, CSS, JS)
app.use(express.static( dirname )) // Redirect root
link to index.html app.get("/", (req, res) => {
res.sendFile(path.join( dirname, "index.html"))

})

// Insert student data app.post("/addStudent", (req, res) => { const
{name,email,dob,department,phone} = req.body const sql = "INSERT INTO
students(name,email,dob,department,phone) VALUES (?,?,?,?,?)"
db.query(sql,[name,email,dob,department,phone] ,(err,result)=>{
if(err){ res.send(err) }else{
res.send("Student Added
Successfully") }

})

})

// Get student data
app.get("/students", (req, res)=>{
db.query("SELECT * FROM
students", (err,result)=>{
if(err){
res.send(err)

}else{
res.json(result)

}

})

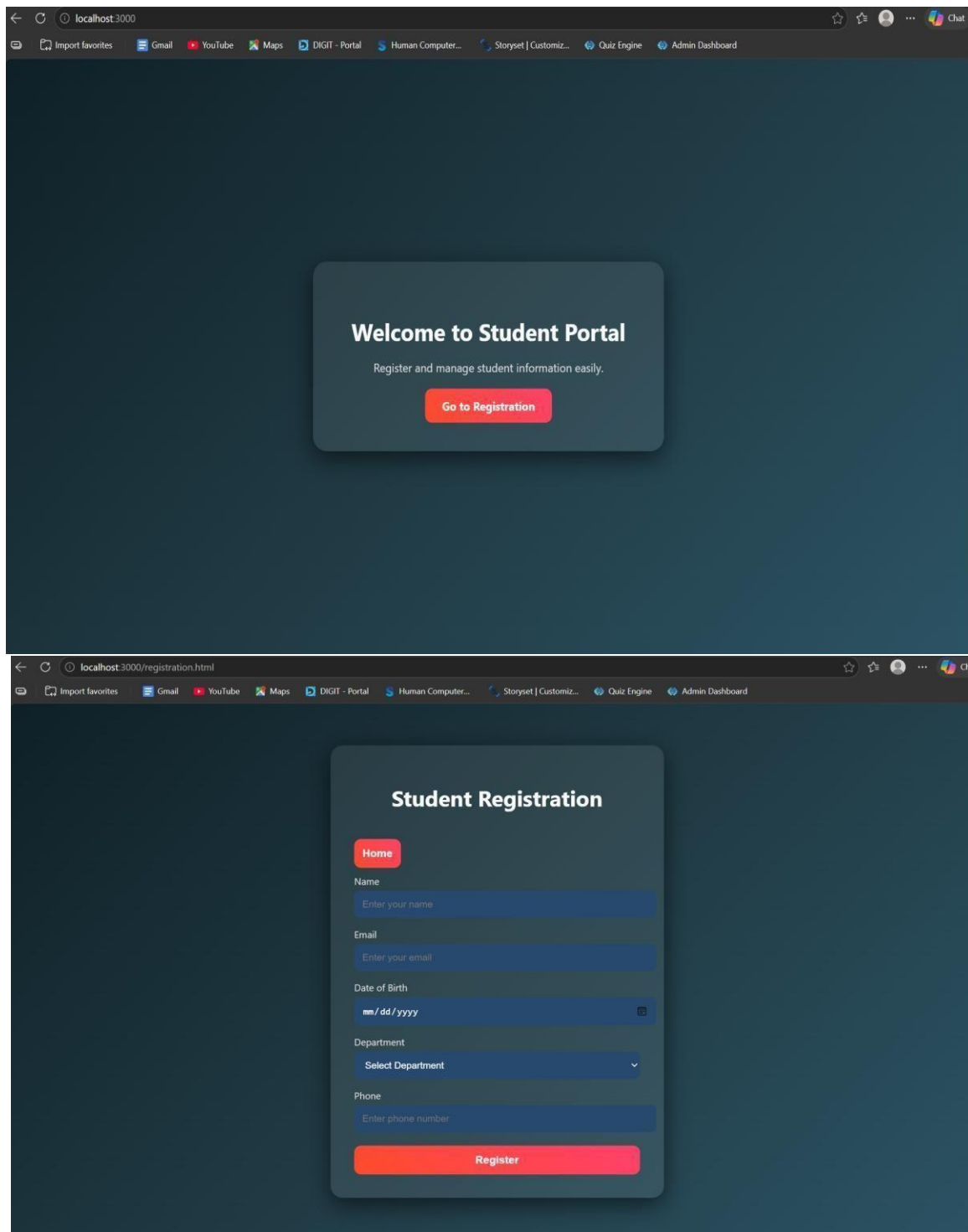
```

```
})  
//      Start      server      app.listen(3000,()=>{  
console.log("Server      running      at:  
http://localhost:3000")  
})
```

DB.JS:

```
const mysql = require("mysql") const  
db = mysql.createConnection({ host:  
"localhost", user: "root", password:  
"fahi",  
  
database: "studentdb"  
})  
db.connect((err)=>{  
if(err){      console.log("Database  
Error:", err)  
}else{      console.log("MySQL      Connected  
Successfully")  
}  
})  
module.exports = db
```

OUTPUTS:



RESULT: The student registration system was successfully developed to store and manage student data using a web interface and MySQL database.

TASK2: Data Retrieval & Sorting Dashboard

AIM: To develop a web-based dashboard that allows users to store, retrieve, and display student data efficiently from a database, and present it in a structured format for easy access and management.

ALGORITHM:

1. Start the application
2. Initialize the server using Express framework
3. Establish database connection using MySQL
4. Load frontend pages (HTML, CSS, JS) in browser

► Data Insertion Process:

5. User enters student details (Name, Email, DOB, Department, Phone)
6. Send data to backend using POST request (/addStudent)
7. Execute SQL query to insert data into database
8. Display success message after insertion

► Data Retrieval Process:

9. Send GET request to server (/students)
10. Fetch all student records from database
11. Convert data into JSON format
12. Display data dynamically in dashboard table.
13. Arrange data in structured format (table/grid)
14. Allow better readability and management
15. Stop.

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

<title>Student Portal</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="home-container">

<h1>Welcome to Student Portal</h1>

<p>Register and manage student information easily.</p>

<a href="registration.html" class="start-btn">Go to Registration</a>

<a href="dashboard.html" class="start-btn">View Dashboard</a>

</div>

</body>

</html>
```

REGISTRATION.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Student Registration</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

<h1>Student Registration</h1>

<button onclick="goHome()" class="home-btn">Home</button>

<form id="studentForm">
```

```
<label>Name</label>
<input type="text" id="name" placeholder="Enter your name" required>
<label>Email</label>
<input type="email" id="email" placeholder="Enter your email" required>
<label>Date of Birth</label>
<input type="date" id="dob" required>
<label>Department</label>
<select id="department" required>
<option value="">Select Department</option>
<option>CSE</option>
<option>IT</option>
<option>ECE</option>
<option>MECH</option>
</select>
<label>Phone</label>
<input type="text" id="phone" placeholder="Enter phone number" required>
<button type="submit">Register</button>
</form>
</div>
<script src="script.js"></script>
</body>
</html>
```

DASHBOARD.HTML:

```
<!DOCTYPE html>
<html>
```



```
<head>

<title>Student Dashboard</title>

<link rel="stylesheet" href="dashboard.css">

</head>

<body>

<div class="container">

<button onclick="goHome()" class="home-btn">Home</button>

<h1>Student Dashboard</h1>

<div class="controls">

<select id="sortSelect">

<option value="">Sort By</option>

<option value="name">Name</option>

<option value="dob">Date of Birth</option>

</select>

<select id="departmentFilter">

<option value="">Filter by Department</option>

<option value="CSE">CSE</option>

<option value="IT">IT</option>

<option value="ECE">ECE</option>

<option value="MECH">MECH</option>

</select>

<button onclick="loadStudents()">Apply</button>

</div>

<table>

<thead>

<tr>
```

```
<th>ID</th>
<th>Name</th>
<th>Email</th>
<th>DOB</th>
<th>Department</th>
<th>Phone</th>
</tr>
</thead>
<tbody id="studentTable"></tbody>
</table>
<h2>Department Count</h2>
<div id="deptCount"></div>
</div>
<script src="dashboard.js"></script>
<script>    function    goHome(){
window.location.href          =
"index.html"
}
</script>
</body>
</html>
```

STYLE.CSS: body{ font-family: 'Segoe UI', sans-serif; background: linear-gradient(135deg,#0f2027,#203a43,#2c5364); height:100vh; display:flex; justify-content:center; align-items:center; margin:0;

```
}
```

```
.container{ width:420px; padding:35px;  
background:rgba(255,255,255,0.08);  
border-radius:18px;    backdrop-filter:  
blur(10px); box-shadow:0 10px 35px  
rgba(0,0,0,0.6);  
}
```

```
h1{ text-align:center;  
color:white; margin-  
bottom:25px;  
}
```

```
h2{ text-  
align:center;  
color:white; margin-  
top:30px; }
```

```
form{ display:flex;  
flex-direction:column;  
}
```

```
label{ color:#ddd;  
font-size:14px;
```

```
margin-bottom:5px;
margin-top:10px;
}
```

```
input,select{
width:100%;
padding:12px;
border:none; border-
radius:8px;
background:#27496d;
color:white; font-
size:14px;
outline:none; margin-
bottom:5px;
}
```

```
input:focus,select:foc
us{ box-shadow:0 0
8px #00eaff;
background:#2c5d8a;
}
```

```
button{ margin-top:15px; padding:12px;
border:none; border-radius:10px;
background:linear-
gradient(45deg,#ff4b2b,#ff416c); color:white; font-
```

```
size:16px; font-weight:bold; cursor:pointer;
transition:0.3s;
}
```

```
button:hover{
transform:scale(1.05); box-
shadow:0 0 15px #ff416c;
}
```

```
#studentList{ margin-
top:15px; text-
align:center;
color:white; font-
size:15px;
}
```

```
.home-container{      text-align:center;
color:white;
background:rgba(255,255,255,0.08);
padding:50px;      border-radius:20px;
backdrop-filter:    blur(10px);      box-
shadow:0 10px 35px rgba(0,0,0,0.6);
width:420px;
}
```

```
.home-container h1{  
font-size:32px;  
margin-bottom:15px;  
}
```

```
.home-container p{  
margin-bottom:30px;  
color:#ddd;  
}
```

```
.start-btn{ text-  
decoration:none;  
padding:12px 25px;  
border-radius:10px;  
background:linear-  
gradient(45deg,#ff4b  
2b,#ff416c);  
color:white; font-  
weight:bold;  
transition:0.3s;  
}
```

```
.start-btn:hover{  
transform:scale(1.05); box-  
shadow:0 0 15px #ff416c;
```

```
}
```

```
DASHBOARD.CSS: body{ font-family:Segoe UI;  
background:linear-  
gradient(135deg,#0f2027,#203a43,#2c5364); margin:0;  
padding:0; display:flex; justify-content:center;  
}
```

```
.container{  
width:900px; margin-  
top:40px;  
background:rgba(255,  
255,255,0.08);  
padding:30px;  
border-radius:15px;  
backdrop-  
filter:blur(10px);  
color:white;  
}
```

```
h1{ text-align:center;  
margin-  
bottom:20px;  
}
```

```
.controls{ display:flex;
gap:10px; margin-
bottom:20px;
}
```

```
select,button{
padding:10px;
border:none; border-
radius:6px;
}
```

```
button{
background:#ff416c;
color:white;
cursor:pointer;
}
```

```
table{ width:100%; border-
collapse:collapse;
background:rgba(255,255,255,0.1)
;
}
```

```
th,td{ padding:10px;
text-align:center;
```



```
}
```

```
th{
```

```
background:#1f4068;
```

```
}
```

```
tr:nth-child(even){
```

```
background:rgba(255,255,255,0.05);
```

```
}
```

```
#deptCount{ margin-
```

```
top:15px; font-
```

```
size:16px;
```

```
}
```

```
.home-btn{
```

```
padding:8px 16px;
```

```
border:none; border-
```

```
radius:6px;
```

```
background:#00c6ff;
```

```
color:white;
```

```
cursor:pointer;
```

```
margin-bottom:10px;
```

```
}
```

SCRIPT.JS:

```

const form = document.getElementById("studentForm") form.addEventListener("submit",
async (e) => {
e.preventDefault()    const    student    =    {    name:
document.getElementById("name").value,                email:
document.getElementById("email").value,                dob:
document.getElementById("dob").value,                department:
document.getElementById("department").value,    phone:
document.getElementById("phone").value
}
try{ const res = await
fetch("http://localhost:3000/addStudent",{ method:"POST",
headers:{
"Content-Type":"application/json"
},
body:JSON.stringify(student)
})
const data = await
res.text() alert(data) //
clear form form.reset()
}catch(error){
console.log("Error:",error)
}
})
function                                goHome(){
window.location.href="index.html"
}

```

SERVER.JS:

```
const express = require("express")
const cors = require("cors") const
path = require("path") const db =
require("./db") const app =
express() app.use(cors())
app.use(express.json()) // Serve
static files (HTML, CSS, JS)
app.use(express.static(__dirname))
// Redirect root link to index.html
app.get("/", (req, res) => {
res.sendFile(path.join(__dirname,
"index.html"))
})
// Insert student data app.post("/addStudent", (req, res) => { const
{name,email,dob,department,phone} = req.body const sql = "INSERT INTO
students(name,email,dob,department,phone) VALUES (?, ?, ?, ?, ?)"
db.query(sql,[name,email,dob,department,phone] ,(err,result)=>{
if(err){ res.send(err) }else{
res.send("Student Added
Successfully")
}
})
```

```

})
//      Get      student      data
app.get("/students",(req,res)=>{ db.query("SELECT
* FROM students",(err,result)=>{
if(err){
res.send(err)
}else{
res.json(result)
}
})
})
//      Start      server      app.listen(3000,()=>{
console.log("Server      running      at:
http://localhost:3000")
})

```

DB.JS:

```

const mysql = require("mysql")
const db = mysql.createConnection({
host: "localhost", user: "root",
password: "fahi", database:
"studentdb"
})
db.connect((err)=>{
if(err){ console.log("Database
Error:", err)

```

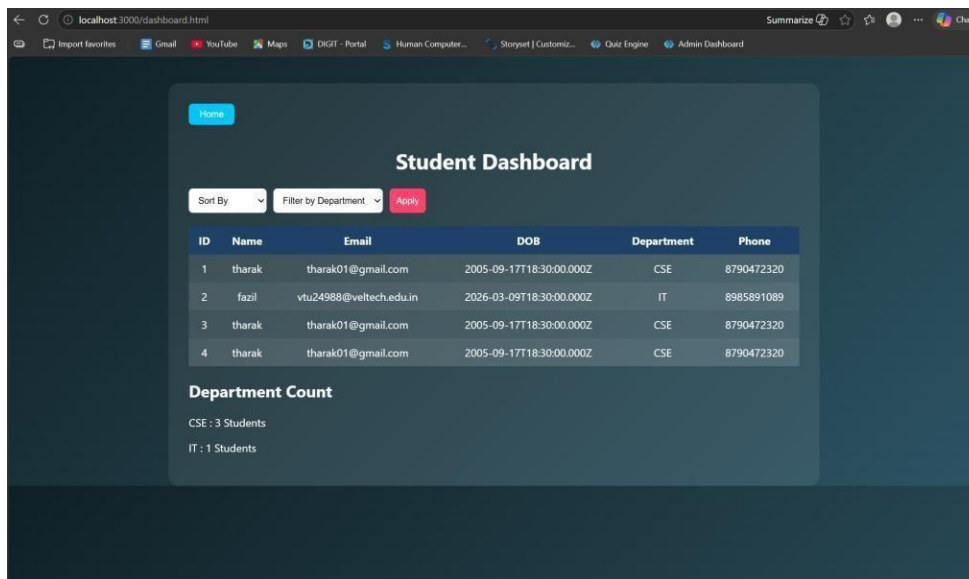
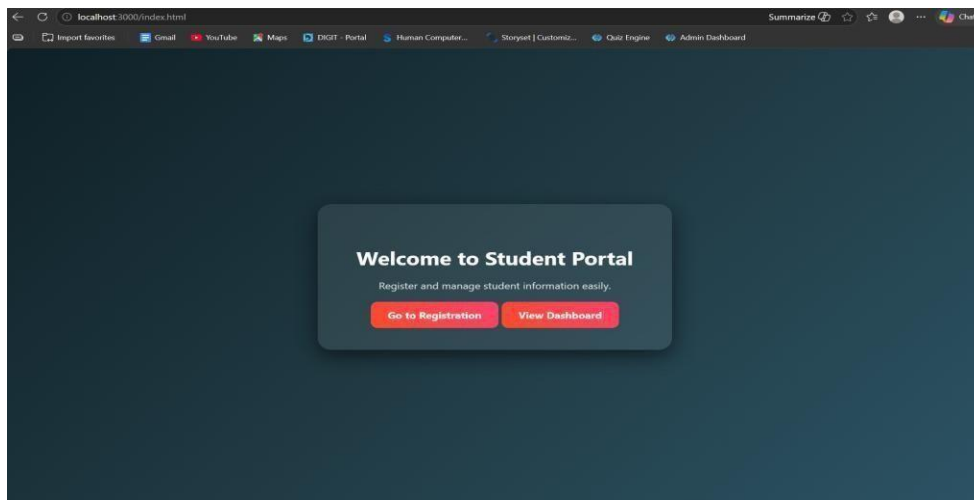
```

}else{    console.log("MySQL    Connected
Successfully")
}
})

module.exports = db

```

OUTPUTS:



RESULT: The system successfully stores, retrieves, and displays student data efficiently through a user-friendly dashboard.

TASK3: : Login System with Validation

AIM: To develop a secure login system that validates user credentials (username and password) and allows access only to authorized users.

ALGORITHM:

1. Start the application
2. Display login form (Username & Password fields)
3. User enters login credentials
4. Perform input validation:
 - Check if fields are empty
 - Validate email format (if applicable)
 - Ensure password meets required criteria
5. Send login data to backend server
6. Server receives request and processes input
7. Compare entered credentials with database records
8. If credentials match:
 - Grant access and redirect to dashboard
9. Else:
 - Display error message ("Invalid Username/Password")
10. End process

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

<title>Student Portal</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="home-container">

<h1>Welcome to Student Portal</h1>

<p>Register and manage student information easily.</p>

<a href="registration.html" class="start-btn">Go to Registration</a>

<a href="dashboard.html" class="start-btn">View Dashboard</a>

</div>

</body>

</html>
```

DASHBOARD.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Student Dashboard</title>

<link rel="stylesheet" href="dashboard.css">

</head>

<body>

<div class="container">

<button onclick="goHome()" class="home-btn">Home</button>

<h1>Student Dashboard</h1>

<div class="controls">
```

```
<select id="sortSelect">
<option value="">Sort By</option>
<option value="name">Name</option>
<option value="dob">Date of Birth</option>
</select>

<select id="departmentFilter">
<option value="">Filter by Department</option>
<option value="CSE">CSE</option>
<option value="IT">IT</option>
<option value="ECE">ECE</option>
<option value="MECH">MECH</option>
</select>

<button onclick="loadStudents()">Apply</button>

</div>

<table>
<thead>
<tr>
<th>ID</th>
<th>Name</th>
<th>Email</th>
<th>DOB</th>
<th>Department</th>
<th>Phone</th>
</tr>
</thead>
<tbody id="studentTable"></tbody>
```



```
</table>

<h2>Department Count</h2>

<div id="deptCount"></div>

</div>

<script src="dashboard.js"></script>

<script>    function    goHome(){

window.location.href = "index.html"

}

</script>

</body>

</html>
```

LOGIN.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Login</title>

<link rel="stylesheet" href="login.css">

</head>

<body>

<div class="login-container">

<h1>Login</h1>

<form id="loginForm">

<label>Username</label>

<input type="text" id="username"> <label>Password</label>

<input type="password" id="password">
```

```
<p id="errorMsg"></p>
<button type="submit">Login</button>
</form>
</div>
<script src="login.js"></script>
</body>
</html>
```

REGISTRATION.HTML:

```
<!DOCTYPE html>
<html>
<head>
<title>Student Registration</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h1>Student Registration</h1>
<button onclick="goHome()" class="home-btn">Home</button>
<form id="studentForm">
<label>Name</label>
<input type="text" id="name" placeholder="Enter your name" required>
<label>Email</label>
<input type="email" id="email" placeholder="Enter your email" required> <label>Date of
Birth</label>
<input type="date" id="dob" required>
```

```
<label>Department</label>
<select id="department" required>
<option value="">Select Department</option>
<option>CSE</option>
<option>IT</option>
<option>ECE</option>
<option>MECH</option>
</select>
<label>Phone</label>
<input type="text" id="phone" placeholder="Enter phone number" required>
<button type="submit">Register</button>
</form>
</div>
<script src="script.js"></script>
</body>
</html>
```

```
STYLE.CSS: body{ font-family: 'Segoe UI', sans-serif; background:
linear-gradient(135deg,#0f2027,#203a43,#2c5364);
height:100vh; display:flex; justify-content:center;

align-items:center;
margin:0;

}
```

```
.container{ width:420px; padding:35px;
background:rgba(255,255,255,0.08);
border-radius:18px;    backdrop-filter:
blur(10px); box-shadow:0 10px 35px
rgba(0,0,0,0.6);
}
```

```
h1{ text-align:center;
color:white; margin-
bottom:25px;
}
```

```
h2{ text-
align:center;
color:white;
margin-top:30px;
}
```

```
form{
display:flex;          flex-
direction:column;
}
```

```
label{ color:#ddd;
font-size:14px;
margin-
```

```
bottom:5px;
margin-top:10px;
}
```

```
input,select{
width:100%;
padding:12px;
border:none; border-
radius:8px;
background:#27496d;
color:white; font-
size:14px; outline:none;
margin-bottom:5px;
}
```

```
input:focus,select:focus{ box-
shadow:0 0 8px #00eaff;
background:#2c5d8a;
}
```

```
button{ margin-top:15px; padding:12px;
border:none; border-radius:10px;
background:linear-
gradient(45deg,#ff4b2b,#ff416c); color:white; font-
```

```
size:16px; font-weight:bold; cursor:pointer;
transition:0.3s;
}
```

```
button:hover{
transform:scale(1.05); box-
shadow:0 0 15px #ff416c;
}
```

```
#studentList{
margin-top:15px;
text-align:center;
color:white; font-
size:15px;
}
.home-container{      text-align:center;
color:white;
background:rgba(255,255,255,0.08);
padding:50px;      border-radius:20px;
backdrop-filter:    blur(10px);    box-
shadow:0 10px 35px  rgba(0,0,0,0.6);
width:420px;
}
```

```
.home-container h1{ font-
size:32px; margin-bottom:15px;
```

```
}
```

```
.home-container p{ margin-  
bottom:30px; color:#ddd;  
}
```

```
.start-btn{ text-decoration:none; padding:12px  
25px; border-radius:10px; background:linear-  
gradient(45deg,#ff4b2b,#ff416c); color:white; font-  
weight:bold; transition:0.3s;  
}
```

```
.start-btn:hover{  
transform:scale(1.05); box-  
shadow:0 0 15px #ff416c;  
}
```

SERVER.JS:

```
const express = require("express") const cors =  
require("cors") const path = require("path") const  
db = require("./db") const app = express()  
app.use(cors()) app.use(express.json())  
app.use(express.urlencoded({ extended: true })) //  
Disable default index.html loading  
app.use(express.static(_dirname, { index: false })))
```

```

// =====
// OPEN LOGIN PAGE FIRST //
=====

app.get("/", (req, res) => {
  res.sendFile(path.join(__dirname, "login.html"))
})

// =====
// LOGIN SYSTEM
// ===== app.post("/login",(req,res)=>{ const
{username,password} = req.body const sql = "SELECT * FROM users
WHERE          username=?          AND          password=?"
db.query(sql,[username,password] ,(err,result)=>{
  if(err){
    res.send("error")
    return
  }
  if(result.length          >          0){
    res.send("success")
  }else{
    res.send("fail")
  }
})
})

// =====
// STUDENT REGISTRATION

```



```

// ===== app.post("/addStudent",(req,res)=>{    const
{name,email,dob,department,phone} = req.body    const    sql    = "INSERT    INTO
students(name,email,dob,department,phone)            VALUES            (?, ?, ?, ?)"
db.query(sql,[name,email,dob,department,phone] ,(err,result)=>{

if(err){ res.send(err) }else{
res.send("Student Added
Successfully")
}

})

})

// =====
// GET STUDENTS (Dashboard)
// =====

app.get("/students",(req,res)=>{

db.query("SELECT * FROM students", (err,result)=>{

if(err){
res.send(err)
}else{
res.json(result)
}
}
}

```

```
}}
```

```
}}
```

```
// =====
```

```
// START SERVER
```

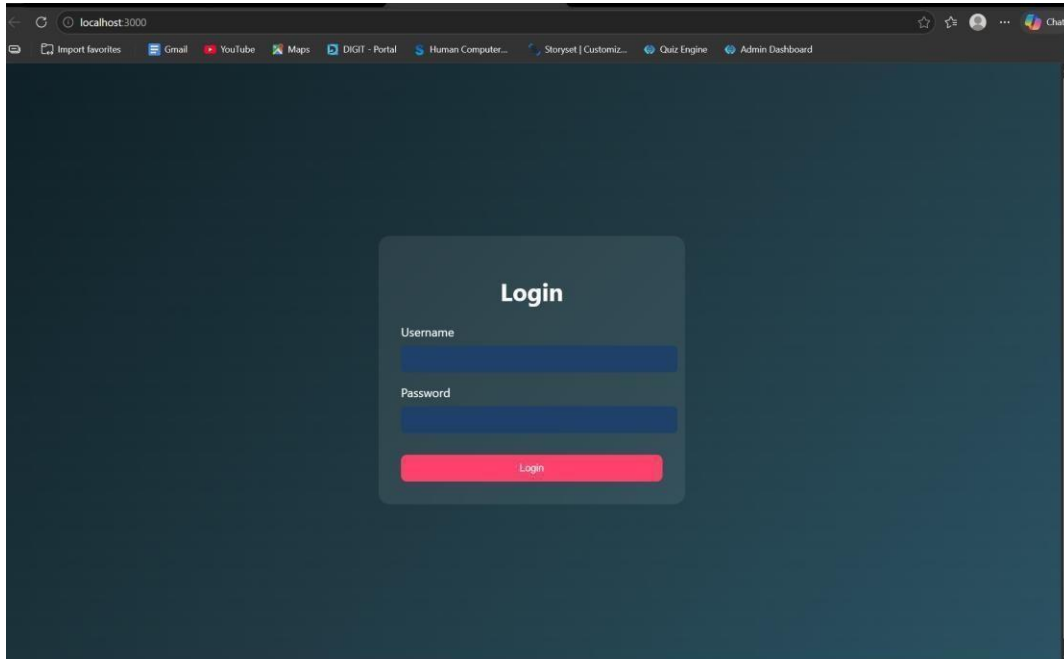
```
// =====
```

```
app.listen(3000,()=>{ console.log("Server running at:
```

```
http://localhost:3000)
```

```
}}
```

OUTPUTS:



RESULT: The system successfully validates user credentials and grants access only to authorized users while preventing invalid logins.

TASK4: Order Management using Joins

AIM: To design and implement an order management system using SQL joins and subqueries to efficiently retrieve and analyze customer, product, and order data.

ALGORITHM:

1. Start the process
2. Create database tables:
 - Customers (CustomerID, Name, etc.)
 - Products (ProductID, ProductName, Price)
 - Orders (OrderID, CustomerID, ProductID, Quantity, TotalAmount)
3. Insert sample data into all tables
- Data Retrieval using Joins:
 4. Use SQL JOIN queries to combine data from Customers, Orders, and Products
 5. Display customer order history with details (Name, Product, Quantity, Total Amount)
- Subquery for Analysis:
 6. Use subquery to find the highest value order
 7. Use subquery with GROUP BY to find the most active customer (maximum number of orders)
- Sorting:
 8. Apply ORDER BY clause to sort results (e.g., highest order value first)
 9. Display results in structured format (table/UI with CSS if frontend used)
 10. Stop

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Login</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="login">

<h2>Admin Login</h2>

<input type="text" id="username" placeholder="Username">

<input type="password" id="password" placeholder="Password">

<button onclick="login()">Login</button>

</div>

<script src="script.js"></script>

</body>

</html>
```

ADMINDASHBOARD.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Order Dashboard</title>

<link rel="stylesheet" href="style.css">

</head>
```

```
<body>

<!-- NAVBAR -->

<div class="navbar">

<h2>Order Management</h2>

<a href="index.html">

<button class="logout-btn">Logout</button>

</a>

</div>

<h1>Order Management Dashboard</h1>

<!-- CARDS SECTION -->

<div class="grid">

<!-- Add Customer -->

<div class="card">

<h3>Add Customer</h3>

<input id="cname" placeholder="Customer Name">

<input id="cemail" placeholder="Email">

<button onclick="addCustomer()">Add Customer</button>

</div>

<!-- Add Product -->

<div class="card">

<h3>Add Product</h3>

<input id="pname" placeholder="Product Name">

<input id="pprice" placeholder="Price">

<button onclick="addProduct()">Add Product</button>

</div>

<!-- Place Order -->
```

```
<div class="card">
  <h3>Place Order</h3>
  <select id="customerSelect">
    <option value="">Select Customer</option>
  </select>
  <select id="productSelect">
    <option value="">Select Product</option>
  </select>
  <input id="qty" placeholder="Quantity">
  <button onclick="placeOrder()">Place Order</button>
</div>
</div>
<!-- ACTION BUTTONS -->
<div class="actions">
  <button onclick="loadOrders()">Order History</button>
  <button onclick="highestOrder()">Highest Order</button>
  <button onclick="activeCustomer()">Most Active Customer</button>
</div>
<!-- RESULT AREA -->
<div id="result"></div>
<script src="script.js"></script>
</body>
</html>
```

STYLE.CSS:

```
*{ margin:0; padding:0; box-sizing:border-box; font-family:
'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
}
```

```
body{ background: linear-
gradient(135deg,#5f6caf,#9a6bff); min-
height:100vh; display:flex; flex-direction:column;
align-items:center;
}
```

```
/* LOGIN PAGE */
```

```
.login{ width:350px; padding:40px;
margin-top:120px; background:white;
border-radius:15px; box-shadow:0 10px
25px rgba(0,0,0,0.2); text-align:center;
}
.login h2{ margin-
bottom:20px;
color:#333;
}
```

```
.login      input{
width:100%;
padding:12px;
margin:10px      0;
```

```
border:1px      solid
#ccc;          border-
radius:8px;
outline:none;
}
```

```
.login button{
width:100%;
padding:12px;
border:none;
background:#5f6caf;
color:white; font-
weight:bold; border-
radius:8px;
cursor:pointer;
transition:0.3s;
}
.login button:hover{ background:#7c4dff;
}
```

```
/* DASHBOARD TITLE */
```

```
h1{ margin-top:30px;
margin-
bottom:25px;
color:white;
```



```
}
```

```
/* CARD GRID */
```

```
.grid{ display:flex;  
gap:25px; flex-  
wrap:wrap; justify-  
content:center;  
margin-bottom:20px;  
}
```

```
/* CARDS */
```

```
.card{  
width:260px; padding:22px; border-  
radius:15px; box-shadow:0 8px 20px  
rgba(0,0,0,0.2); text-align:center;  
transition:0.3s; color:white;  
}
```

```
/* Different card colors */
```

```
.card:nth-child(1){  
background:linear-  
gradient(135deg,#ff7a18,#ffb347);  
}
```

```
.card:nth-child(2){ background:linear-  
gradient(135deg,#00c6ff,#0072ff);  
}
```

```
.card:nth-child(3){ background:linear-  
gradient(135deg,#43e97b,#38f9d7); color:#222;  
}
```

```
.card:hover{ transform:translateY(-  
6px); }
```

```
.card h3{ margin-  
bottom:10px;  
}
```

```
/* INPUTS */
```

```
.card input{  
width:90%;  
padding:10px;  
margin:6px; border-  
radius:6px;  
border:none;  
}
```

```
/* CARD BUTTON */
```

```
.card button{  
padding:10px 15px;  
border:none;  
background:white;  
color:#333; border-  
radius:6px;  
cursor:pointer; font-  
weight:bold;  
transition:0.3s;  
  
}
```

```
.card button:hover{ background:#f1f1f1;  
  
}
```

```
/* ACTION BUTTONS */
```

```
button{ margin:10px;  
padding:12px 18px;  
border:none;  
border-radius:6px;  
background:#ffffff;
```

```
color:#333; font-  
weight:bold;  
cursor:pointer;  
transition:0.3s;  
}
```

```
button:hover{  
background:#eaeaea;  
}
```

```
/* TABLE */  
table{ width:85%; margin:auto;  
margin-top:30px; border-  
collapse:collapse; background:white;  
border-radius:12px; overflow:hidden;  
box-shadow:0 5px 15px  
rgba(0,0,0,0.2);  
}
```

```
th{  
background:#5f6caf;  
color:white;  
padding:14px; font-  
size:18px;  
}
```

```
td{ padding:12px; text-align:center; border-bottom:1px solid #ddd; color:#222; font-weight:500; }
tr:nth-child(even){ background:#f7f7f7; }
```

```
tr:hover{ background:#eef1ff; }
```

```
/* RESULT TEXT */
```

```
#result{ margin-top:20px; color:white; font-size:18px; text-align:center; }
```

```
/* NAVBAR */
```

```
.navbar{
```

```
width:100%;
```

```
display:flex; justify-  
content:space-between; align-  
items:center; padding:15px  
40px; background:#2c2f5c;  
color:white;  
box-shadow:0 4px 10px rgba(0,0,0,0.2);  
  
}
```

```
.navbar h2{ font-  
weight:600;  
}
```

```
.logout-btn{  
  
background:#ff5252;  
color:white;  
border:none;  
padding:10px 16px;  
border-radius:6px;  
cursor:pointer;  
font-weight:bold;  
  
}
```

```
.logout-btn:hover{

background:#ff1744;
}

.actions{

display:flex; justify-
content:center;
gap:20px; margin-
top:30px; flex-
wrap:wrap;

}

.actions button{
padding:12px 20px;
font-size:16px;

border:none;

border-radius:8px;
background:#5f6caf
; color:white;
cursor:pointer;
transition:0.3s;

}

.actions button:hover{
background:#7c4dff;
transform:scale(1.05);
```

```
}
```

SCRIPT.JS:

```
// LOGIN FUNCTION
```

```
function login(){
```

```
const username=document.getElementById("username").value
```

```
const password=document.getElementById("password").value
```

```
fetch("/login",{
```

```
method:"POST",
```

```
headers:{"Content-Type":"application/json"},
```

```
body:JSON.stringify({username,password})
```

```
})
```

```
.then(res=>res.json())
```

```
.then(data=>{
```

```
if(data.success){
```



```
window.location="admindashboard.html"
}else{ alert("Invalid
login")} }

})

}

// ADD CUSTOMER

function addCustomer(){

fetch("/add-customer",{

method:"POST",

headers:{"Content-Type":"application/json"},

body:JSON.stringify({

name:document.getElementById("cname").value,
email:document.getElementById("cemail").value

})

})

})
```

```
.then(res=>res.text())
```

```
.then(alert)
```

```
}
```

```
// ADD PRODUCT
```

```
function addProduct(){
```

```
  fetch("/add-product",{
```

```
    method:"POST",
```

```
    headers:{"Content-Type":"application/json"},
```

```
    body:JSON.stringify({
```

```
      product:document.getElementById("pname").value,
```

```
      price:document.getElementById("pprice").value
```

```
    })
```

```
  })
```

```
.then(res=>res.text())
```

```
.then(alert)
```

```
}
```

```
// LOAD CUSTOMERS INTO DROPDOWN
```

```
async function loadCustomers(){
```

```
const res = await fetch("/customers") const
```

```
data = await res.json()
```

```
let select=document.getElementById("customerSelect")
```

```
if(!select) return
```

```
select.innerHTML=<option value="">Select Customer</option>
```

```
data.forEach(c=>{
```

```
select.innerHTML+=`<option value="${c.id}">${c.name}</option>`
```

```
})
```

```
}
```

```
// LOAD PRODUCTS INTO DROPDOWN
```

```
async function loadProducts(){
```

```
const res = await fetch("/products")
```

```
const data = await res.json()
```

```
let select=document.getElementById("productSelect")
```

```
if(!select) return
```

```
select.innerHTML='<option value="">Select Product</option>'
```

```
data.forEach(p=>{
```

```
select.innerHTML+=`<option value="${p.id}">${p.product_name}</option>`
```

```
})
```

```
}
```

```
// PLACE ORDER
```

```
function placeOrder(){
```

```
fetch("/place-order",{

method:"POST", headers:{"Content-

Type":"application/json"}, body:JSON.stringify({

customer_id:document.getElementById("customerSelect").value,

product_id:document.getElementById("productSelect").value,

quantity:document.getElementById("qty").value

})

})

.then(res=>res.text())

.then(alert)

}
```

```
// ORDER HISTORY (JOIN)
```

```
async function loadOrders(){
```

```
const res=await fetch("/orders") const
```

```
data=await res.json()
```

```
let
```

```
html=<table><tr><th>Customer</th><th>Product</th><th>Price</th><th>Qty</th><th>Total</th></tr>
```

```
data.forEach(o=>{
```

```
html+=`
```

```
<tr>
```

```
<td>${o.name}</td>
```

```
<td>${o.product_name}</td>
```

```
<td>${o.price}</td>
```

```
<td>${o.quantity}</td>
```

```
<td>${o.total}</td>
```

```
</tr>
```

```
,
```

```
})
```

```
html+=</table>
```

```
document.getElementById("result").innerHTML=html
```

```
}
```

```
// HIGHEST ORDER (SUBQUERY)
```

```
async function highestOrder(){
```

```
const res=await fetch("/highest-order")
```

```
const data=await res.json()
```

```
document.getElementById("result").innerHTML=
```

```
`<h2>Highest Order</h2>
```

```
${data[0].name} - ₹${data[0].total}`
```

```
}
```

```
// MOST ACTIVE CUSTOMER
```

```
async function activeCustomer(){
```

```
const res=await fetch("/active-customer")
```

```
const data=await res.json()
```

```
document.getElementById("result").innerHTML=
```

```
`<h2>Most Active Customer</h2> ${data[0].name} (${data[0].total_orders} orders)`
```

```
}
```

```
// AUTO LOAD DROPDOWNS WHEN PAGE LOADS
```

```
window.onload=function(){
```

```
loadCustomers()
```

```
loadProducts()
```

```
}
```

SERVER.JS:

```
const express=require("express")
```

```
const db=require("./db")
```

```
const app=express()
```



```
app.use(express.json())
```

```
app.use(express.static(_dirname))
```

```
// LOGIN
```

```
app.post("/login",(req,res)=>{
```

```
const {username,password}=req.body
```

```
const sql="SELECT * FROM users WHERE username=? AND password=?"
```

```
db.query(sql,[username,password] ,(err,result)=>{
```

```
if(err) throw err
```

```
if(result.length>0){
```

```
res.json({success:true})
```

```
}else{
```

```
res.json({success:false})
```

```
}
```

```
})
```

```
})
```

```
// ADD CUSTOMER
```

```
app.post("/add-customer",(req,res)=>{
```

```
  const {name,email}=req.body
```

```
  const sql="INSERT INTO customers(name,email) VALUES(?,?)"
```

```
  db.query(sql,[name,email] ,(err)=>{
```

```
    if(err) throw err
```

```
    res.send("Customer Added Successfully")
```

```
  })
```

```
})
```

```
// ADD PRODUCT
```

```
app.post("/add-product",(req,res)=>{
```

```
  const {product,price}=req.body
```

```
  const sql="INSERT INTO products(product_name,price) VALUES(?,?)"
```

```
db.query(sql,[product,price],(err)=>{
```

```
if(err) throw err
```

```
res.send("Product Added Successfully")  
})
```

```
})
```

```
// GET CUSTOMERS (for dropdown)
```

```
app.get("/customers",(req,res)=>{
```

```
const sql="SELECT * FROM customers"
```

```
db.query(sql,(err,result)=>{
```

```
if(err) throw err
```

```
res.json(result)
```

```
})
```

```
})
```

```
// GET PRODUCTS (for dropdown)
```

```
app.get("/products",(req,res)=>{  
  const sql="SELECT * FROM products"
```

```
  db.query(sql,(err,result)=>{
```

```
    if(err) throw err
```

```
    res.json(result)
```

```
  })
```

```
})
```

```
// PLACE ORDER
```

```
app.post("/place-order",(req,res)=>{
```

```
  const {customer_id,product_id,quantity}=req.body
```

```
  const sql="INSERT INTO orders(customer_id,product_id,quantity) VALUES(?,?,?)"
```

```
db.query(sql,[customer_id,product_id,quantity]},{err)=>{
```

```
if(err) throw err
```

```
res.send("Order Placed Successfully")  
})
```

```
})
```

```
// ORDER HISTORY (JOIN)
```

```
app.get("/orders",(req,res)=>{
```

```
const sql=`
```

```
SELECT customers.name,  
products.product_name,  
products.price,  
orders.quantity,  
(products.price * orders.quantity) AS total
```

```
FROM orders
```

```
JOIN customers
```

```
ON orders.customer_id = customers.id
```

JOIN products

ON orders.product_id = products.id

ORDER BY orders.order_date DESC

,

db.query(sql,(err,result)=>{

if(err) throw err

res.json(result)

})

})

// HIGHEST VALUE ORDER (SUBQUERY)

app.get("/highest-order",(req,res)=>{

const sql=`

SELECT customers.name,

(products.price * orders.quantity) AS total

FROM orders

JOIN customers

ON orders.customer_id = customers.id

JOIN products

ON orders.product_id = products.id WHERE (products.price * orders.quantity) =

(
SELECT MAX(products.price * orders.quantity)

FROM orders

JOIN products

ON orders.product_id = products.id

)

,

db.query(sql,(err,result)=>{

if(err) throw err

res.json(result)

```
}}
```

```
}}
```

```
// MOST ACTIVE CUSTOMER
```

```
app.get("/active-customer",(req,res)=>{
```

```
const sql=`
```

```
SELECT customers.name,
```

```
COUNT(orders.id) AS total_orders
```

```
FROM customers
```

```
JOIN orders
```

```
ON customers.id = orders.customer_id
```

```
GROUP BY customers.id
```

```
ORDER BY total_orders DESC
```

```
LIMIT 1
```

```
,
```



```
db.query(sql,(err,result)=>{
```

```
if(err) throw err
```

```
res.json(result)
```

```
})
```

```
})
```

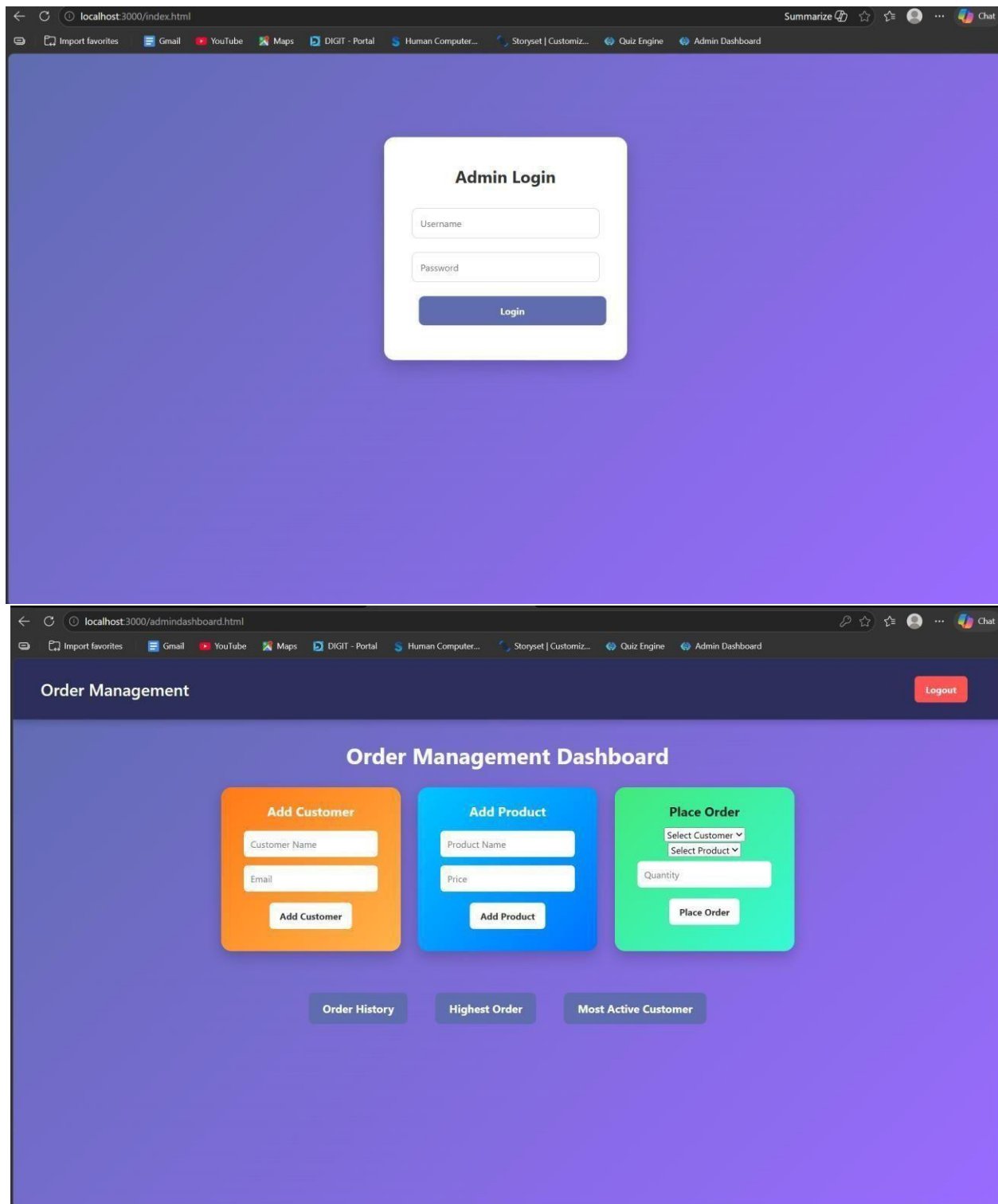
```
// START SERVER
```

```
app.listen(3000,()=>{
```

```
console.log("Server running at http://localhost:3000")
```

```
})
```

OUTPUTS:



RESULT: The system successfully retrieves and analyzes customer order data using joins and subqueries, displaying meaningful insights like highest order value and most active customer.

TASK5: Transaction-Based Payment Simulation

AIM: To simulate an online payment system using database transactions that ensures safe and reliable fund transfer between user and merchant accounts using COMMIT and ROLLBACK operations.

ALGORITHM:

1. Start the process
2. Create database tables:
 - User Account (UserID, Balance)
 - Merchant Account (MerchantID, Balance)
3. Insert initial balance values for user and merchant
- Transaction Process:
 4. Begin transaction (START TRANSACTION)
 5. Deduct amount from user account
 6. Add amount to merchant account
- Validation:
 7. Check if user has sufficient balance
 - If yes → proceed
 - If no → go to rollback
- Commit / Rollback:
 8. If all operations are successful → COMMIT transaction
 9. If any error occurs → ROLLBACK transaction
 10. Display transaction status (Success / Failed)
11. Stop

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

<title>Payment Simulation</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

<h2>Online Payment</h2>

<select id="user" onchange="updateBalances()"></select>

<select id="merchant" onchange="updateBalances()"></select>

<input type="number" id="amount" placeholder="Enter amount">

<button onclick="pay()">Pay</button>

<h3 id="userBalance"></h3>

<h3 id="merchantBalance"></h3>

<p id="result"></p>

</div>

<script src="script.js"></script>

</body>

</html>
```

STYLE.CSS: body{
font-family: Arial;
background:
linear-
gradient(135deg,

```
#dfe9f3,#ffffff);
```

```
display:flex;
```

```
justify-
```

```
content:center;
```

```
align-
```

```
items:center;
```

```
height:100vh;
```

```
}
```

```
.container{ background:white;
```

```
padding:30px; border-radius:15px; box-
```

```
shadow:0 10px 25px rgba(0,0,0,0.2);
```

```
text-align:center; width:400px;
```

```
}
```

```
select, input{
```

```
padding:10px;
```

```
margin:10px;
```

```
width:90%; border-
```

```
radius:5px;
```

```
border:1px solid #ccc;
```

```
}
```

```
button{ padding:10px;
```

```
width:95%;
```

```
background:green;
color:white;
border:none; border-
radius:5px;
cursor:pointer; font-
size:16px;
}
```

```
button:hover{
background:darkgreen;
}
```

```
h3{
margin:5px;
color:#333;
}
```

```
SCRIPT.JS:      let
usersData  = [] let
merchantsData = []
function  loadData(){
fetch("/balances")
.then(res=>res.json())
.then(data=>{
usersData =
```

```

data.users
merchantsData =
data.merchants let
userSelect =
document.getElemen
tById("user") let
merchantSelect =
document.getElemen
tById("merchant")
userSelect.innerHTML
=""
merchantSelect.inner
HTML=""

// fill users
usersData.forEach(u=>{    userSelect.innerHTML    +=    `<option
value="${u.id}">${u.name}</option>`

})
window.onload = loadData // fill merchants merchantsData.forEach(m=>{
merchantSelect.innerHTML    +=    `<option
value="${m.id}">${m.name}</option>`

})
//    show    initial    balances
updateBalances()

})
}

```

```

function updateBalances(){
    let userId =
    document.getElementById("user").value
    let merchantId =
    document.getElementById("merchant").value
    let user =
    usersData.find(u=>u.id == userId)
    let merchant =
    merchantsData.find(m=>m.id == merchantId)
    document.getElementById("userBalance").innerText =
    "User Balance: ₹" + user.balance
    document.getElementById("merchantBalance").innerText = "Shop Balance: ₹" +
    merchant.balance
}

function pay(){
    let userId =
    document.getElementById("user").value
    let merchantId =
    document.getElementById("merchant").value
    let amount =
    document.getElementById("amount").value
    fetch("/pay",{
    method:"POST", headers:{
    "Content-Type":"application/json"
    },
    body:JSON.stringify({
    userId:userId,
    merchantId:merchantId,
    amount:amount
    })
    })
}

```



```

.then(res=>res.json()) .then(data=>{
document.getElementById("result").innerText =
data.message loadData()
})
}

```

window.onload = loadData

SERVER.JS: const express =

require("express") const db =

require("./db") const app = express()

app.use(express.json())

app.use(express.static(_dirname))

app.get("/", (req, res) => { res.sendFile(
dirname + "/index.html")

})

```

app.post("/pay", (req, res) => { const { userId,
merchantId, amount } = req.body const amt =
parseInt(amount) if (!amt || amt <= 0) { return
res.json({ message: "Enter valid amount" })
}

```

```

db.beginTransaction(err => { if (err) return
res.status(500).send("Transaction error")

```

```

// 1. Get user balance db.query("SELECT balance FROM users WHERE id=?",
[userId], (err, result) => { if (err || result.length === 0) { return db.rollback(() =>
res.send("User not found"))
}

```

```

}

```

```
let balance = result[0].balance if (balance < amt)
{ return db.rollback(() => { res.json({ message:
  "Insufficient Balance" })

  })
}
```

```
// 2. Deduct from user db.query(
  "UPDATE users SET balance = balance - ? WHERE id=?",
  [amt, userId],
  (err) => {

    if (err) {
      return db.rollback(() => res.send("Deduction failed"))
    }

    // 3. Add to merchant db.query(
      "UPDATE merchants SET balance = balance + ? WHERE id=?",
      [amt, merchantId],
      (err) => {

        if (err) {
          return db.rollback(() => res.send("Merchant update failed"))
        }

        // 4. Commit
        db.commit(err => {
```

```

        if (err) {
            return db.rollback(() => res.send("Commit failed"))
        }

        res.json({ message: "Payment Successful  " })
    })
}
)
}
)
})
})
})
app.get("/balances", (req, res) => {

```

```

    db.query("SELECT * FROM users", (err, users) => {

```

```

        if (err) return res.send("Error fetching users")

```

```

        db.query("SELECT * FROM merchants", (err, merchants) => {

```

```

            if (err) return res.send("Error fetching merchants")

```

```

            res.json({ users: users,
                merchants: merchants

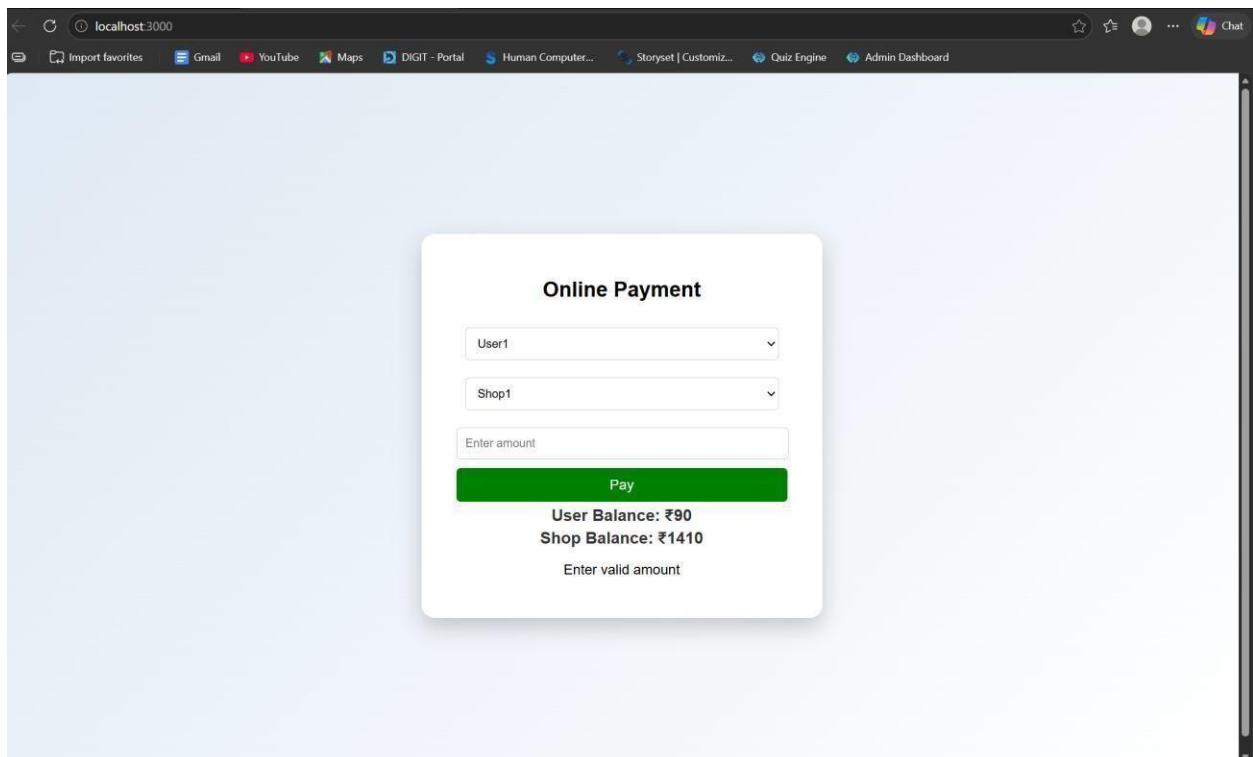
```

```

    })
  })
})
app.listen(3000, () => { console.log("Server running on
http://localhost:3000")
})

```

OUTPUTS:



The screenshot shows a web browser window with the address bar set to 'localhost:3000'. The browser's address bar and tabs are visible at the top. The main content area displays a light blue background with a white modal form titled 'Online Payment'. The form contains three input fields: a dropdown menu showing 'User1', another dropdown menu showing 'Shop1', and a text input field labeled 'Enter amount'. Below these fields is a prominent green button labeled 'Pay'. Underneath the 'Pay' button, the current balances are displayed: 'User Balance: ₹90' and 'Shop Balance: ₹1410'. At the bottom of the form, there is a message 'Enter valid amount'.

RESULT: The system successfully performs secure payment transactions by updating balances accurately using COMMIT on success and ROLLBACK on failure.

TASK6: Automated Logging using Triggers & Views

AIM: To develop a web-based application for automated logging of database activities using MySQL triggers and views, integrated with a Node.js backend and frontend interface, in order to track INSERT and UPDATE operations and generate daily activity reports.

ALGORITHM:

1. Start the program
2. Create database and required tables
3. Create triggers for INSERT and UPDATE operations
4. Store actions in log table automatically
5. Create a view for daily activity report
6. Develop backend using Node.js and Express
7. Connect backend with MySQL database
8. Create APIs for add, update, logs, and report
9. Design frontend using HTML, CSS, and JavaScript
10. Run the application and display results
11. Stop the program

PROGRAM:**INDEX.HTML:**

```
<!DOCTYPE html>

<html>

<head>

  <title>Employee Audit System</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<h2>Employee Management</h2>

<div class="form">
```

```
<input type="number" id="emp_id" placeholder="Employee ID">
<input type="text" id="name" placeholder="Name">
<input type="number" id="salary" placeholder="Salary">
<button onclick="addEmployee()">Add</button>
</div>
<div class="form">
  <input type="number" id="update_id" placeholder="Employee ID">
  <input type="number" id="new_salary" placeholder="New Salary">
  <button onclick="updateEmployee()">Update</button>
</div>
<button onclick="loadLogs()">View Logs</button>
<button onclick="loadReport()">View Daily Report</button>
<h3>Output</h3>
<pre id="output"></pre>
<script src="script.js"></script>
</body>
</html>
```

```
STYLE.CSS: body { font-
family: Arial; text-align:
center;    background:
#f4f4f4;

}
.form { margin:
    15px;

}
```

```

input      {
    padding:
    8px; margin:
    5px;
}
button { padding:
    8px      15px;
    background: blue;
    color:    white;
    border:   none;
    cursor: pointer;
}
button:hover { background:
    darkblue;
}
pre { background:
    white; padding:
    10px; width: 60%;
    margin: auto;
    text-align: left;
}

```

SCRIPT.JS:

```

const BASE_URL = "http://localhost:3000"; // Add Employee function addEmployee() { const data
=      {      emp_id:      document.getElementById("emp_id").value,      name:
document.getElementById("name").value, salary: document.getElementById("salary").value
    };

```

```

    fetch(`${BASE_URL}/add`, { method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify(data)
    })

    .then(res => res.text())

    .then(data => alert(data));
  }

  // Update Employee function updateEmployee() { const id =
  document.getElementById("update_id").value; const salary =
  document.getElementById("new_salary").value;

```

```

    fetch(`${BASE_URL}/update/${id}`, { method:
      "PUT", headers: { "Content-Type":
      "application/json" }, body: JSON.stringify({
      salary })
    })

    .then(res => res.text())

    .then(data => alert(data));
  }

  // Load Logs function
  loadLogs() {
    fetch(`${BASE_URL}/logs`)

    .then(res => res.json())

    .then(data => {
      document.getElementById("output").innerText =
      JSON.stringify(data, null, 2);
    });
  }

```



```

    });
}
// Load Report function
loadReport() {
  fetch(`${BASE_URL}/report`)
    .then(res => res.json())
    .then(data => {
      document.getElementById("output").innerText =
        JSON.stringify(data, null, 2);
    });
}

```

DB.JS:

```

const mysql = require('mysql2');
const db = mysql.createConnection({
  host: 'localhost', user: 'root',
  password: 'fahi',

  database: 'audit_db'
});
db.connect((err) => {
  if (err) {
    console.error('DB Connection Failed:', err);
  } else {
    console.log('Connected to MySQL');
  }
});

```

```
module.exports = db;
```

OUTPUTS:

Employee Management

Employee ID Name Salary [Add](#)

Employee ID New Salary [Update](#)

[View Logs](#) [View Daily Report](#)

Output

```
[
  {
    "log_id": 1,
    "emp_id": 1,
    "action_type": "INSERT",
    "action_time": "2026-04-13T04:59:54.000Z"
  },
  {
    "log_id": 2,
    "emp_id": 1,
    "action_type": "UPDATE",
    "action_time": "2026-04-13T05:00:36.000Z"
  }
]
```

RESULT: The system successfully logs all INSERT and UPDATE operations automatically using triggers and displays daily activity reports through a web interface.

TASK7: Interactive Web Form with Events & Functions

AIM: To develop an interactive web-based feedback form with real-time validation, UI effects, and backend data storage using HTML, CSS, JavaScript, and Node.js.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design feedback form with input fields (name, email, message)
4. Apply CSS styling for layout and visual effects

5. Define reusable validation functions in JavaScript
6. Capture user input using DOM elements
7. Apply keyup event for real-time validation
8. Add hover effect to highlight input fields
9. Implement double-click event on submit button
10. Validate all fields before submission
11. Send data to server using fetch API (POST request)
12. Store data in database module and display success message

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <title>Feedback Form</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2> Customer Feedback</h2>

  <form id="feedbackForm">

    <input type="text" id="name" placeholder="Your Name" required>

    <span class="error" id="nameError"></span>

    <input type="email" id="email" placeholder="Your Email" required>

    <span class="error" id="emailError"></span>
```

```

    <textarea id="message" placeholder="Your Feedback" required></textarea>

    <span class="error" id="msgError"></span>

    <button type="button" id="submitBtn">Submit</button>

</form>

<p id="successMsg"></p>
</div>

<script src="script.js"></script>

</body>
</html>
STYLE.CSS: body { font-family: 'Segoe UI', sans-
serif; background: linear-gradient(135deg, #1f1c2c,
#928dab); display: flex; justify-content: center; align-items:
center; height: 100vh; color: white;

}

.container { background: rgba(255, 255,
255, 0.1); padding: 25px; border-radius:
15px; backdrop-filter: blur(10px); width:
320px; box-shadow: 0 10px 25px
rgba(0,0,0,0.3);

}

h2 { text-align:
center;

}

input, textarea {
width: 100%;
padding: 10px;

```

```
margin: 8px 0;
border-radius:
8px; border: none;
outline: none;
transition: 0.3s;
}
input:hover, textarea:hover {
transform: scale(1.05); box-
shadow: 0 0 8px #00f7ff;
}
button {
width: 100%;
padding: 10px;
background:
#00c6ff; border:
none; border-radius:
10px; color: white;
font-weight: bold;
cursor: pointer;
}
button:hover { background:
#0072ff;
}
.error { color:
#ff6b6b; font-
size: 12px;
```

```

}
#successMsg { text-align: center;
margin-top: 10px;
color: #00ffae;
}

```

SCRIPT.JS:

```

const nameInput = document.getElementById("name"); const
emailInput  =  document.getElementById("email");  const
messageInput = document.getElementById("message"); const
submitBtn    =    document.getElementById("submitBtn");
function validateName() { if (nameInput.value.length < 3) {
document.getElementById("nameError").innerText = "Min 3
characters"; return false;
}
document.getElementById("nameError").innerText    =    "";
return true;
}
function validateEmail() { let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/; if
(!emailInput.value.match(pattern)) {
document.getElementById("emailError").innerText = "Invalid email";
return false;
}
document.getElementById("emailError").innerText    =    "";
return true;
}

```

```

function validateMessage() { if (messageInput.value.length < 5) {
    document.getElementById("msgError").innerText = "Too short";
    return false;
}
document.getElementById("msgError").innerText = "";
return true;
}

nameInput.addEventListener("keyup", validateName); emailInput.addEventListener("keyup",
validateEmail); messageInput.addEventListener("keyup", validateMessage);
submitBtn.addEventListener("dblclick", () => { if (validateName() && validateEmail() &&
validateMessage()) { sendData();
    } else { alert("Fix errors before
    submitting!");
    }
});

function sendData() {
    fetch("http://localhost:3000/submit",
    { method: "POST", headers: {
        "Content-Type": "application/json"
    },
    body: JSON.stringify({ name:
        nameInput.value, email:
        emailInput.value, message:
        messageInput.value
    })
    })
}

```

```

    .then(res => res.json())
    .then(data => { document.getElementById("successMsg").innerText = "    Feedback
        Submitted!";
    });
}

```

```

SERVER.JS: const express = require("express");
const cors = require("cors"); const bodyParser =
require("body-parser");    const    path    =
require("path"); const db = require("./db"); const
app    =    express();    app.use(cors());
app.use(bodyParser.json()); app.use(express.static(
dirname));  app.get("/", (req, res) => {
res.sendFile(path.join(_dirname, "index.html"));
});
app.post("/submit", (req, res) => { const {
    name, email, message } = req.body;
    db.saveFeedback({ name, email, message });
    res.json({ status: "success" });
});
app.listen(3000, () => { console.log("Server running on
http://localhost:3000/");
});

```

DB.JS:

```

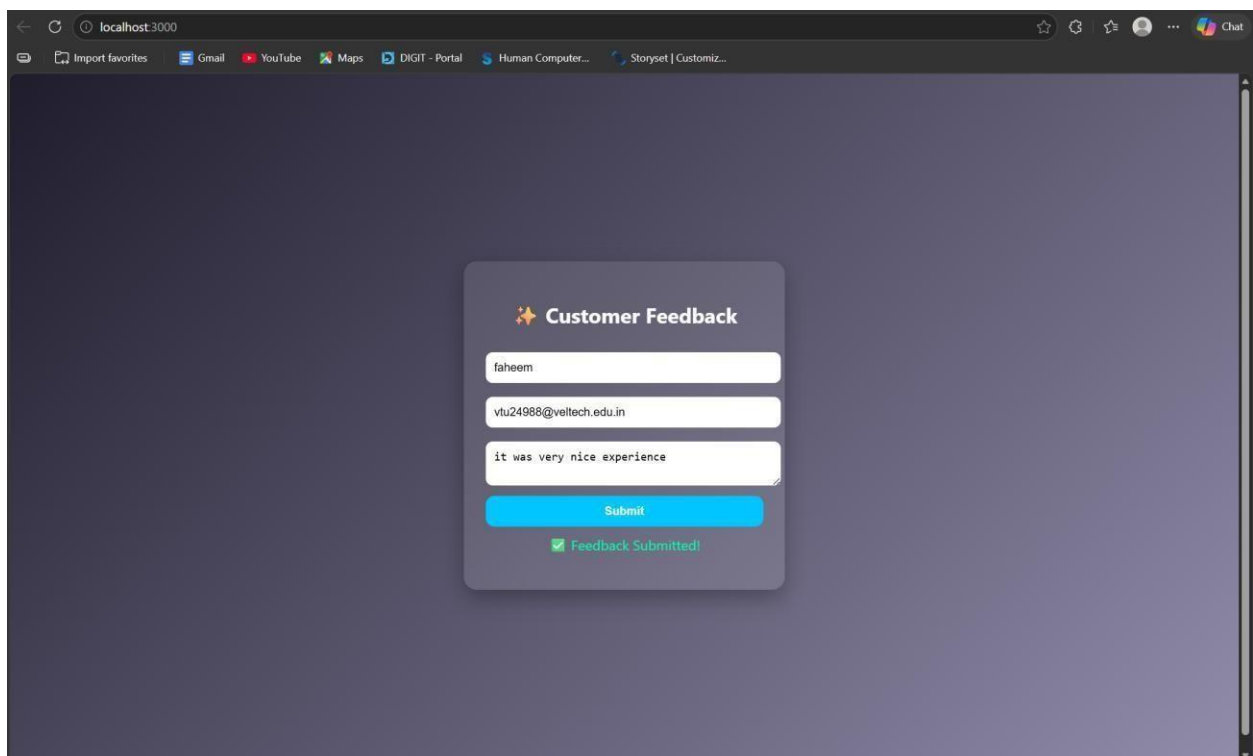
let feedbacks = []; function
saveFeedback(data)    {

```



```
feedbacks.push(data);  
console.log("Saved:", data);  
  
}  
  
module.exports      =      {  
    saveFeedback  
  
};
```

OUTPUT:



RESULT: The interactive feedback form was successfully implemented with real-time validation, enhanced UI effects, and backend data handling.

TASK8: Employee Management

AIM: To develop a simple Employee Management system using HTML, CSS, JavaScript, and Node.js for adding and displaying employee data.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design the user interface with input fields (name, role)
4. Apply CSS styling for layout and centering
5. Capture user input using JavaScript
6. Create function to add employee data
7. Send data to server using fetch API (POST request)
8. Create backend using Node.js and Express
9. Receive and store employee data in memory (db.js)
10. Create API to fetch all employee records (GET request)
11. Display employee list dynamically on the webpage
12. Stop the program

PROGRAM:**INDEX.HTML:**

```
<!DOCTYPE html>

<html>

<head>

  <title>Employee Management</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2>Employee Management</h2>
```

```
<input id="name" placeholder="Name">
<input id="role" placeholder="Role">
<button onclick="addEmployee()">Add</button>
<ul id="list"></ul>
</div>
<script src="script.js"></script>
</body>
</html>
```

```
STYLE.CSS: body { font-
family:      Arial;
background:  #1e1e2f;
color:  white;  display:
flex;    justify-content:
center;    align-items:
center;  height: 100vh;
margin: 0;
}
.container { text-align:
center;
}
input  { padding:
10px; margin: 5px;
border-radius:
5px; border: none;
}
```

```
button { padding: 10px;
  background:
    #00c6ff;    border:
    none; border-radius:
    5px; cursor: pointer;
}
```

SCRIPT.JS:

```
function addEmployee() { const name =
  document.getElementById("name").value; const role =
  document.getElementById("role").value;
  fetch("http://localhost:3000/add", { method: "POST",
  headers: {
    "Content-Type": "application/json"
  },
  body: JSON.stringify({ name, role })
})
  .then(res => res.json())
  .then(() => loadEmployees());
}

function loadEmployees() { fetch("http://localhost:3000/all")
  .then(res => res.json())
  .then(data => { let list =
    document.getElementById("list");
    list.innerHTML = "";
```

```

    data.forEach(emp => { let li =
        document.createElement("li");
        li.innerText = emp.name + " - " + emp.role;
        list.appendChild(li);
    });
});
}

loadEmployees();

```

SERVER.JS:

```

const express = require("express");
const cors = require("cors"); const
path = require("path"); const db =
require("./db"); const app =
express(); app.use(cors());

app.use(express.json());
app.use(express.static(_dirname));
app.get("/", (req, res) => {
    res.sendFile(path.join(_dirname,
        "index.html"));
});

app.post("/add", (req, res) => {
    db.addEmployee(req.body);
    res.json({ msg: "Added" });
});

app.get("/all", (req, res) => { res.json(db.getEmployees());

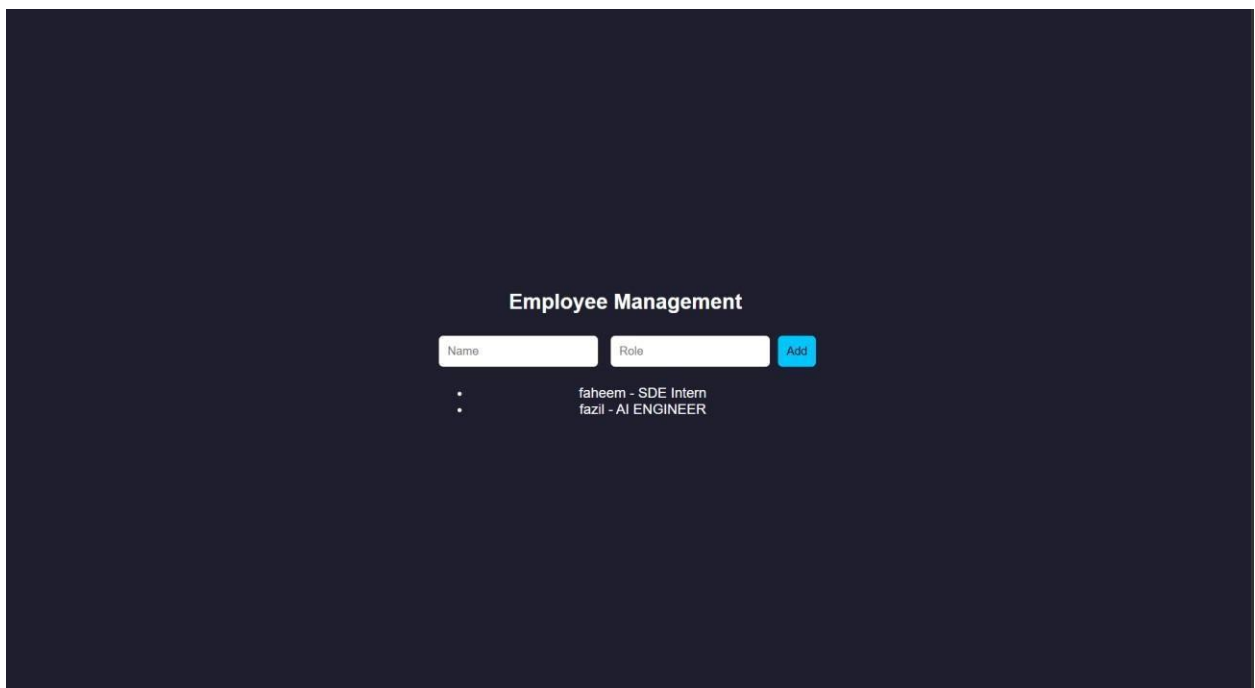
```

```
});  
  
app.listen(3000, () => { console.log("• Server running at:  
http://localhost:3000/");  
});
```

DB.JS:

```
let employees = [];  
function  
addEmployee(emp) {  
  employees.push(emp);  
}  
  
function getEmployees() { return  
  employees;  
}  
  
module.exports = { addEmployee, getEmployees };
```

OUTPUT:



RESULT: The Employee Management system was successfully implemented with a userfriendly interface and dynamic data handling using Node.js backend.

TASK9: Employee MVC APP

AIM: To develop a basic MVC-based Employee application using HTML, CSS, JavaScript, and Node.js to handle user requests and display employee details.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI form to accept employee details
4. Apply CSS styling for better user interface
5. Capture user input using JavaScript
6. Send request to server using fetch API
7. Create controller logic in server.js
8. Define routes to handle requests (/add, /all)
9. Store employee data in memory using db.js
10. Retrieve employee data from database module
11. Send response back to frontend (view)
12. Display employee details dynamically and stop

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>  
  
<html>  
  
<head>
```

```
<title>Employee MVC</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2>Employee Details</h2>

  <input id="name" placeholder="Name">

  <input id="role" placeholder="Role">

  <button onclick="addEmployee()">Add</button>

  <ul id="list"></ul>

</div>

<script src="script.js"></script>

</body>

</html>
```

STYLE.CSS:

```
* {
  margin: 0; padding: 0;
  box-sizing: border-
  box;
}

body { font-family: 'Segoe UI', sans-serif; background:
  linear-gradient(135deg, #1f1c2c, #928dab); height:
  100vh;
```



```
display: flex; justify-
content: center; align-
items: center;

color: white;
}
.container { background: rgba(255, 255,
    255, 0.1); padding: 30px; border-
radius: 15px; backdrop-filter:
blur(10px); width: 320px;

text-align: center;

box-shadow: 0 10px 25px rgba(0,0,0,0.4);
}
h2 { margin-bottom:
    15px;
}
input { width:
    100%;

padding:
10px; margin:
8px 0;

border: none;
border-radius:
8px; outline: none;
```

```
    transition: 0.3s;
}
input:hover,
input:focus { transform:
    scale(1.05); box-shadow: 0 0
    8px #00f7ff;
}
button { width:
    100%;
    padding:
    10px;
    margin-top: 10px;

    background: #00c6ff;
    border: none;
    border-radius: 10px;

    color: white; font-
    weight: bold;
    cursor: pointer;

    transition: 0.3s;
}
button:hover {
    background: #0072ff;
```

```

    transform:
      scale(1.05);
  } ul { list-style: none;
margin-top: 15px;

}
li
{
  background:
    rgba(255,255,255,0.15);    padding:
    8px; margin: 5px 0;
  border-radius: 6px;

  transition: 0.3s;
}
li:hover      {      background:
    rgba(255,255,255,0.3);
}

```

SCRIPT.JS:

```

function  addEmployee()  {  const  name  =
  document.getElementById("name").value; const role =
  document.getElementById("role").value; fetch("/add", {
  method:  "POST",  headers:  {  "Content-Type":
  "application/json" }, body: JSON.stringify({ name, role })
  })
  .then(() => loadEmployees());

```

```

}

function loadEmployees() { fetch("/all")

  .then(res => res.json())

  .then(data => { let list =

    document.getElementById("list");

    list.innerHTML = ""; data.forEach(emp => {

    let li = document.createElement("li");

    li.innerText = emp.name + " - " + emp.role;

    list.appendChild(li);

    });

  });

}

```

loadEmployees();

SERVER.JS:

```

const express = require("express");

const cors = require("cors"); const path

= require("path"); const db =

require("./db"); const app = express();

app.use(cors());

app.use(express.json());

app.use(express.static(____dirname));

app.post("/add", (req, res) => {

db.addEmployee(req.body); res.json({

msg: "Employee Added" });

});

app.get("/all", (req, res) => { res.json(db.getEmployees());

```

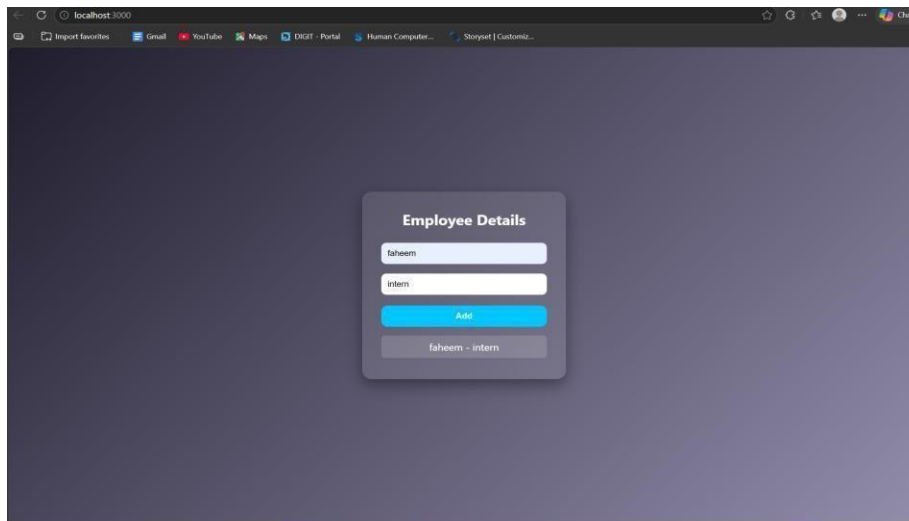
```
});
app.listen(3000,      ()      =>      {
    console.log("http://localhost:3000");
});
```

DB.JS:

```
let employees = []; function
addEmployee(emp)      {
employees.push(emp);

}
function getEmployees() { return
    employees;
}
module.exports = { addEmployee, getEmployees };
```

OUTPUT:



RESULT: The MVC-based Employee application was successfully implemented using Node.js, where user requests were processed and employee details were displayed dynamically.

TASK10: Student Crud App

AIM: To develop a Student CRUD application using HTML, CSS, JavaScript, and Node.js to perform create, read, update, and delete operations.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI for student input (id, name, course)
4. Apply CSS for styling and layout
5. Capture user input using JavaScript
6. Send data to server using fetch API
7. Create backend using Node.js and Express
8. Implement Create operation (POST request)
9. Implement Read operation (GET request)
10. Implement Update operation (PUT request)
11. Implement Delete operation (DELETE request)
12. Display updated student data dynamically and stop

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Student CRUD</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">
```

```

<h2>Student CRUD</h2>

<input id="id" placeholder="ID">

<input id="name" placeholder="Name">

<input id="course" placeholder="Course">

<button onclick="addStudent()">Add</button>

<button onclick="updateStudent()">Update</button>

<ul id="list"></ul>

</div>

<script src="script.js"></script>

</body>

</html>
STYLE.CSS: body { font-family: 'Segoe UI';
background: linear-gradient(135deg,#1f1c2c,#928dab);
height: 100vh; display:flex; justify-content:center; align-
items:center; color:white;

}

.container      {      background:
    rgba(255,255,255,0.1);
padding:20px; border-radius:10px;
text-align:center;

}

input           {
    display:block;
margin:8px
auto;
padding:8px;

```

```

}
button { margin:5px;
padding:8px;
background:#00c6ff
; border:none;
color:white;
}

```

SCRIPT.JS:

```

function addStudent() { let data = getInput();
    fetch("/add", { method: "POST", headers:
    {"Content-Type":"application/json"},    body:
    JSON.stringify(data)
    }).then(loadStudents);
}

function loadStudents() {
    fetch("/all")
    .then(res => res.json())
    .then(data => {
        let list = document.getElementById("list");
        list.innerHTML = ""; data.forEach(s => { let
        li = document.createElement("li");
        li.innerHTML = `
            ${s.id} - ${s.name} (${s.course})
            <button onclick="deleteStudent(${s.id})"> </button>
        `;
        list.appendChild(li);
    });
}

```



```

    });
  });
}
function updateStudent() { let data = getInput();
  fetch("/update", { method: "PUT", headers:
    {"Content-Type":"application/json"},    body:
    JSON.stringify(data)
  }).then(loadStudents);
}
function deleteStudent(id) { fetch(`/delete/${id}`,
  { method: "DELETE" })
  .then(loadStudents);
}
function getInput() { return
  {
    id: document.getElementById("id").value, name:
    document.getElementById("name").value,
    course:
    document.getElementById("course").value
  };
}
loadStudents();

```

SERVER.JS:

```

const express = require("express");
const cors = require("cors"); const

```

```

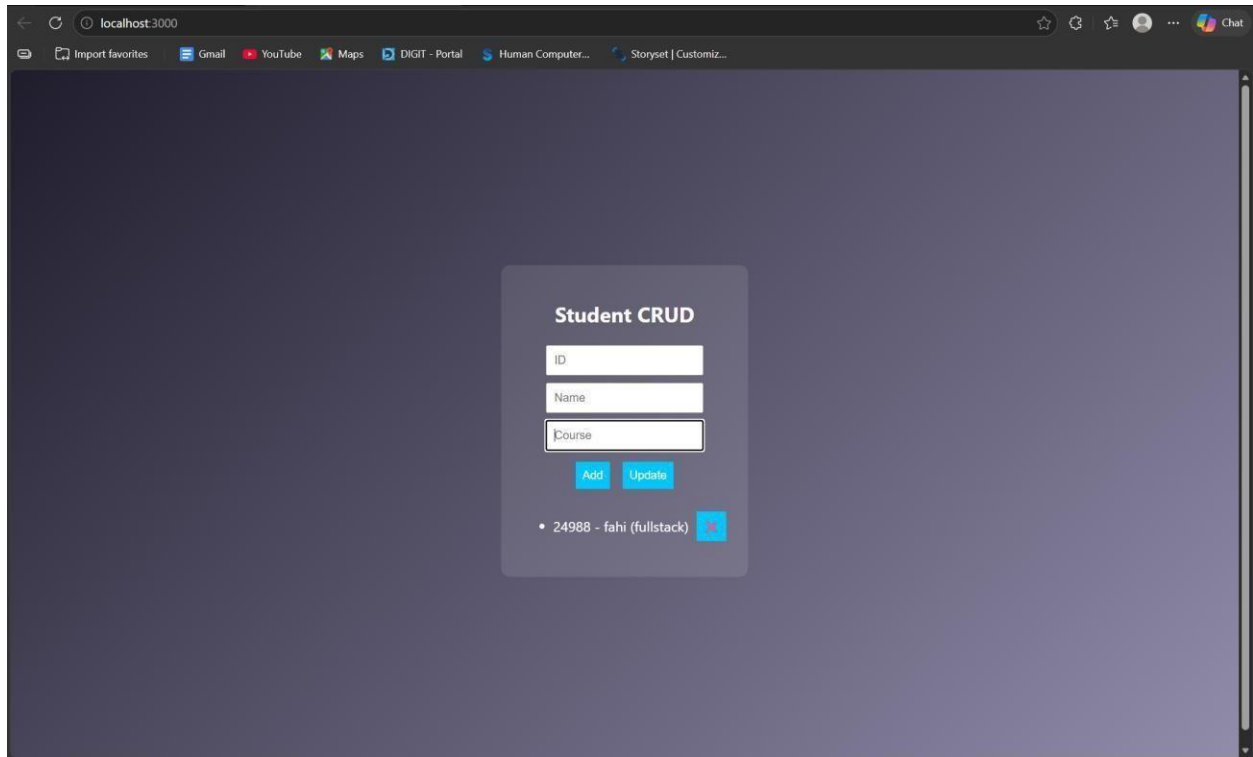
db = require("./db"); const app =
express(); app.use(cors());
app.use(express.json());
app.use(express.static(_dirname));
// CREATE app.post("/add",
(req, res) => {
db.add(req.body);
res.json({msg:"Added"});
});
// READ app.get("/all", (req,
res) => {
res.json(db.getAll());
});
// UPDATE app.put("/update",
(req, res) => {
db.update(req.body);
res.json({msg:"Updated"});
});
// DELETE app.delete("/delete/:id",
(req, res) => {
db.remove(req.params.id);
res.json({msg:"Deleted"});
});
app.listen(3000, () => console.log("http://localhost:3000"));

```

DB.JS:

```
let students = [];  
function add(s) {  
  students.push(s);  
}  
function getAll() { return  
  students;  
}  
function update(updated) { students  
  = students.map(s =>  
    s.id == updated.id ? updated : s  
  );  
}  
function remove(id) { students =  
  students.filter(s => s.id != id);  
}  
module.exports = { add, getAll, update, remove };
```

OUTPUT:



RESULT: The Student CRUD application was successfully implemented with full create, read, update, and delete operations using Node.js and a dynamic web interface.

TASK11: Student Data Layer

AIM: To implement a data access layer for managing student records with filtering, sorting, and pagination using HTML, CSS, JavaScript, and Node.js.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI for student details (id, name, age, department)
4. Apply CSS styling for layout and UI
5. Capture input using JavaScript
6. Send data to backend using fetch API
7. Create Node.js server using Express

8. Store student data in memory (db.js)
9. Implement filtering (by department/age)
10. Implement sorting (by name/age)
11. Implement pagination (limit & page number)
12. Display filtered, sorted, paginated data

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Student Data Layer</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2>Student Data</h2>

  <input id="name" placeholder="Name">

  <input id="age" placeholder="Age">

  <input id="dept" placeholder="Department">

  <button onclick="addStudent()">Add</button>

  <hr>

  <input id="filterDept" placeholder="Filter Dept">

  <input id="sortBy" placeholder="Sort (name/age)">

  <input id="page" placeholder="Page No">

  <button onclick="loadStudents()">Search</button>
```

```
        <ul id="list"></ul>
    </div>
    <script src="script.js"></script>
</body>
</html>    STYLE.CSS:
```

```
body { font-family: Arial;
background:    #1e1e2f;
color:        white;
display:flex;    justify-
content:center;    align-
items:center;
height:100vh;

}
```

```
.container    {    text-
    align:center;
}
```

```
input {
    display:block;
    margin:5px;
    padding:8px;
}
```

SCRIPT.JS:

```
function addStudent() {
    const data = { name:
        val("name"),    age:
```

```

    val("age"),      dept:
    val("dept")

    };

    fetch("/add", { method: "POST", headers:
        {"Content-Type":"application/json"}, body:
        JSON.stringify(data)

    }).then(loadStudents);
}

function loadStudents() { const
    dept = val("filterDept"); const
    sort = val("sortBy"); const
    page    =    val("page");
    fetch(`/students?dept=${dept}&sort=${sort}&page=${page}`)

    .then(res => res.json())
    .then(data => { let list =
        document.getElementById("list");

        list.innerHTML = ""; data.forEach(s => { let li =
            document.createElement("li"); li.innerText =
            `${s.name}    (${s.age})    -    ${s.dept}`;
            list.appendChild(li);

        });

    });
}

```

```
function      val(id){      return
    document.getElementById(id).value;
}
```

```
loadStudents();
```

SERVER.JS: const express =

```
require("express"); const cors =
```

```
require("cors"); const db =
```

```
require("./db"); const app =
```

```
express(); app.use(cors());
```

```
app.use(express.json());
```

```
app.use(express.static(_dirname));
```

```
// Add student app.post("/add",
```

```
(req, res) => { db.add(req.body);
```

```
res.json({msg:"Added"});
```

```
});
```

```
//    Filter    +    Sort    +    Pagination
```

```
app.get("/students", (req, res) => { let {
```

```
dept, sort, page } = req.query; let data =
```

```
db.getAll(); // FILTER if (dept) { data =
```

```
data.filter(s => s.dept === dept);
```

```
}
```

```
// SORT if (sort === "name") {
```

```
data.sort((a,b)=>a.name.localeCompare(b.name));
```

```
} else if (sort === "age") { data.sort((a,b)=>a.age
```

```
- b.age);
```

```
}
```



```
// PAGINATION
let limit = 3; page = parseInt(page) || 1; let
start = (page - 1) * limit; let paginated =
data.slice(start, start + limit);
res.json(paginated);

});

app.listen(3000, () => console.log("http://localhost:3000")); DB.JS:
let students = [];
function add(s) {
students.push(s);
}
function getAll() { return
students;
}
module.exports = { add, getAll };
```

OUTPUT:

Student Data

faheem
21
cse
Add

cse
21
1
Search

- faheem (21) - cse

RESULT: The data access layer was successfully implemented with filtering, sorting, and pagination using Node.js and dynamic frontend interaction.

TASK12: Product Api

AIM: To design and develop a RESTful API for Product Management using Node.js that supports GET, POST, PUT, and DELETE operations with JSON responses.

ALGORITHM:

1. Start the program
2. Create project files (server.js, db.js, HTML, CSS, JS)
3. Design product structure (id, name, price)
4. Create Node.js server using Express
5. Enable JSON parsing using middleware
6. Implement POST method to add products
7. Implement GET method to fetch all products
8. Implement PUT method to update product details

9. Implement DELETE method to remove a product
10. Store product data in memory using db.js
11. Send JSON responses for all API requests
12. Test APIs using browser or REST client

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

    <title>Product API</title>

</head>

<body>

<h2>Product API Test</h2>

<input id="id" placeholder="ID">

<input id="name" placeholder="Name">

<input id="price" placeholder="Price">

<button onclick="add()">Add</button>

<button onclick="get()">Get All</button>

<ul id="list"></ul>

<script src="script.js"></script>

</body>

</html>
```

STYLE.CSS:

```
/* Reset */

* {
```

```

margin: 0; padding: 0;
box-sizing: border-
box;
}
/* Background */ body { font-family: 'Segoe UI', sans-serif;
background: linear-gradient(135deg, #141e30, #243b55);
height: 100vh; display: flex; justify-content: center; align-
items: center; color: white;
}
/* Container */
.container { background: rgba(255, 255,
255, 0.1); padding: 25px; border-
radius: 15px;
backdrop-filter: blur(10px);

width: 350px; text-align: center; box-
shadow: 0 10px 25px rgba(0,0,0,0.4);
}
/* Heading */ h2 {
margin-bottom: 15px;
}
/* Inputs */ input {
width: 100%;
padding: 10px;
margin: 8px 0;
border: none;

```

```
border-radius: 8px;
outline: none;
transition: 0.3s;
}

/* Hover + focus effect */ input:hover,

input:focus { transform:
    scale(1.05); box-shadow: 0 0
    8px #00f7ff;
}

/* Buttons */ button {
width: 100%; padding:
10px; margin-top: 8px;
background: #00c6ff;
border: none; border-
radius: 10px; color:
white; font-weight:
bold; cursor: pointer;
transition: 0.3s;
}

/* Button hover */
button:hover {
background: #0072ff;
transform: scale(1.05);
}

/* Product List */
```

```
ul { list-style: none;
    margin-top:
    15px;
}
/* List items */
li {
    background:
    rgba(255,255,255,0.15); padding:
    10px; margin: 6px 0; border-radius:
    8px; display: flex; justify-content:
    space-between; align-items: center;
    transition: 0.3s;
}
/* Hover effect */ li:hover {
background: rgba(255,255,255,0.3);
}
/* Optional delete button inside list */
li button { width: auto; padding: 5px
8px; font-size: 12px; background:
#ff4b5c;
}
li button:hover { background:
    #ff1e38;
}
```

```

SCRIPT.JS: function add() { fetch("/products", {
method:    "POST",    headers:    {"Content-
Type":"application/json"},          body:
JSON.stringify(getData())

    }).then(get);
}

function get() { fetch("/products")

    .then(res => res.json())
    .then(data    =>    {    let    list    =
        document.getElementById("list"); list.innerHTML
        =    "";    data.forEach(p    =>    {    let    li    =
        document.createElement("li");    li.innerText    =
        `${p.id}    -    ${p.name}    -    ₹${p.price}`;
        list.appendChild(li);

    });

    });
}

function    getData()    {    return    {    id:
    document.getElementById("id").value,    name:
    document.getElementById("name").value,    price:
    document.getElementById("price").value

    };
}

```

SERVER.JS:

```

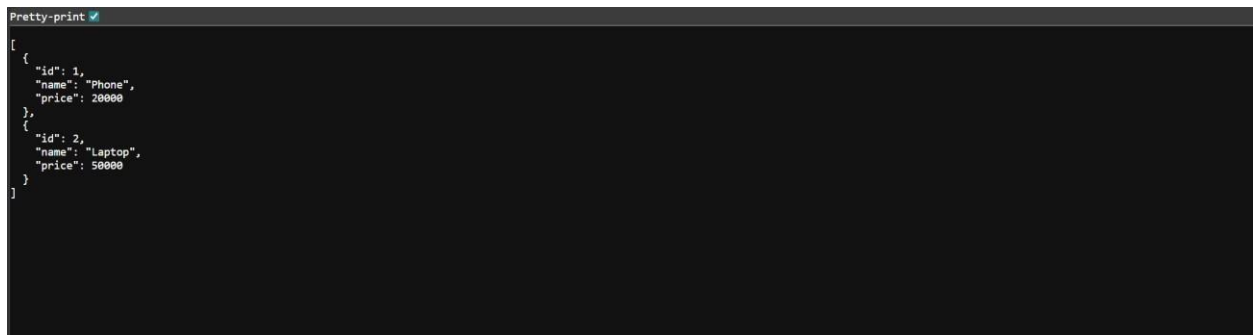
const express = require("express");
const cors = require("cors"); const
db = require("./db"); const app =
express();      app.use(cors());
app.use(express.json()); // GET all
products      app.get("/products",
(req, res) => { res.json(db.getAll());
});
//      POST      add      product
app.post("/products", (req, res) => {
db.add(req.body); res.json({ message:
"Product added" });
});
//      PUT      update      product
app.put("/products/:id", (req, res) => {
db.update(req.params.id,      req.body);
res.json({ message: "Product updated" });
});
// DELETE product app.delete("/products/:id",
(req, res) => { db.remove(req.params.id);
res.json({ message: "Product deleted" });
});
// Server start app.listen(3000, () => { console.log("•   API running
at http://localhost:3000/products");
});

```

DB.JS:


```
let products = [];  
function getAll() {  
  return products;  
}  
  
function add(p) { products.push(p);  
}  
  
function update(id, newData) { products  
  = products.map(p =>  
    p.id == id ? { ...p, ...newData } : p  
  );  
}  
  
function remove(id) { products =  
  products.filter(p => p.id != id);  
}  
  
module.exports = { getAll, add, update, remove };
```

OUTPUT:



```
[  
  {  
    "id": 1,  
    "name": "Phone",  
    "price": 20000  
  },  
  {  
    "id": 2,  
    "name": "Laptop",  
    "price": 50000  
  }  
]
```

RESULT: The RESTful API for Product Management was successfully implemented with GET, POST, PUT, and DELETE methods returning JSON responses.

TASK13: PRODUCT-API

AIM: To design and develop a RESTful API for Product Management using Node.js that supports GET, POST, PUT, and DELETE operations with JSON responses.

ALGORITHM:

1. Start the program
2. Create project files (server.js, db.js, HTML, CSS, JS)
3. Design product structure (id, name, price)
4. Create Node.js server using Express
5. Enable JSON parsing using middleware
6. Implement POST method to add products
7. Implement GET method to fetch all products
8. Implement PUT method to update product details
9. Implement DELETE method to remove a product
10. Store product data in memory using db.js
11. Send JSON responses for all API requests
12. Test APIs using browser or REST client

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

    <title>Product API</title>

</head>

<body>

<h2>Product API Test</h2>
```

```
<input id="id" placeholder="ID">
<input id="name" placeholder="Name">
<input id="price" placeholder="Price">
<button onclick="add()">Add</button>
<button onclick="get()">Get All</button>
<ul id="list"></ul>
<script src="script.js"></script>
</body>
</html>
```

STYLE.CSS:

```
/* Reset */
* {
  margin: 0; padding: 0;
  box-sizing: border-
  box;
}
/* Background */ body { font-family: 'Segoe UI', sans-serif;
background: linear-gradient(135deg, #141e30, #243b55);
height: 100vh; display: flex; justify-content: center; align-
items: center; color: white;
}
/* Container */
.container { background: rgba(255, 255,
  255, 0.1); padding: 25px; border-
  radius: 15px;
  backdrop-filter: blur(10px);
```

```
width: 350px; text-align: center; box-
shadow: 0 10px 25px rgba(0,0,0,0.4);
}
/* Heading */ h2 {
margin-bottom: 15px;
}
/* Inputs */ input {
width:      100%;
padding:    10px;
margin:     8px 0;
border:     none;
border-radius: 8px;
outline:    none;
transition: 0.3s;
}
/* Hover + focus effect */ input:hover,
input:focus { transform:
scale(1.05); box-shadow: 0 0
8px #00f7ff;
}
/* Buttons */ button {
width: 100%; padding:
10px; margin-top: 8px;
background: #00c6ff;
```

```
border: none; border-
radius: 10px; color:
white; font-weight:
bold; cursor: pointer;
transition: 0.3s;
}
/* Button hover */
button:hover {
background: #0072ff;
transform: scale(1.05);
}
/* Product List */
ul { list-style: none;
margin-top:
15px;
}
/* List items */
li {
background:
rgba(255,255,255,0.15); padding:
10px; margin: 6px 0; border-radius:
8px; display: flex; justify-content:
space-between; align-items: center;
transition: 0.3s;
}
```

```

/* Hover effect */ li:hover {
background: rgba(255,255,255,0.3);
}
/* Optional delete button inside list */
li button { width: auto; padding: 5px
8px; font-size: 12px; background:
#ff4b5c;
}
li button:hover { background:
#ff1e38;
}
SCRIPT.JS: function add() { fetch("/products", {
method: "POST", headers: {"Content-
Type":"application/json"}, body:
JSON.stringify(getData())
}).then(get);
}
function get() { fetch("/products")
.then(res => res.json())
.then(data => { let list =
document.getElementById("list"); list.innerHTML
= ""; data.forEach(p => { let li =
document.createElement("li"); li.innerText =
`${p.id} - ${p.name} - ₹${p.price}`;
list.appendChild(li);
});
});

```

```

    });
}
function    getData()    {    return    {    id:
    document.getElementById("id").value,    name:
    document.getElementById("name").value,    price:
    document.getElementById("price").value

    };
}

```

SERVER.JS:

```

const express = require("express");
const cors = require("cors"); const
db = require("./db"); const app =
express();    app.use(cors());
app.use(express.json()); // GET all
products    app.get("/products",
(req, res) => { res.json(db.getAll());
});
//    POST    add    product
app.post("/products", (req, res) => {
db.add(req.body); res.json({ message:
"Product added" });
});
//    PUT    update    product
app.put("/products/:id", (req, res) => {

```

```

db.update(req.params.id, req.body);
res.json({ message: "Product updated" });
});
// DELETE product app.delete("/products/:id",
(req, res) => { db.remove(req.params.id);
res.json({ message: "Product deleted" });
});
// Server start app.listen(3000, () => { console.log("• API running
at http://localhost:3000/products");
});

```

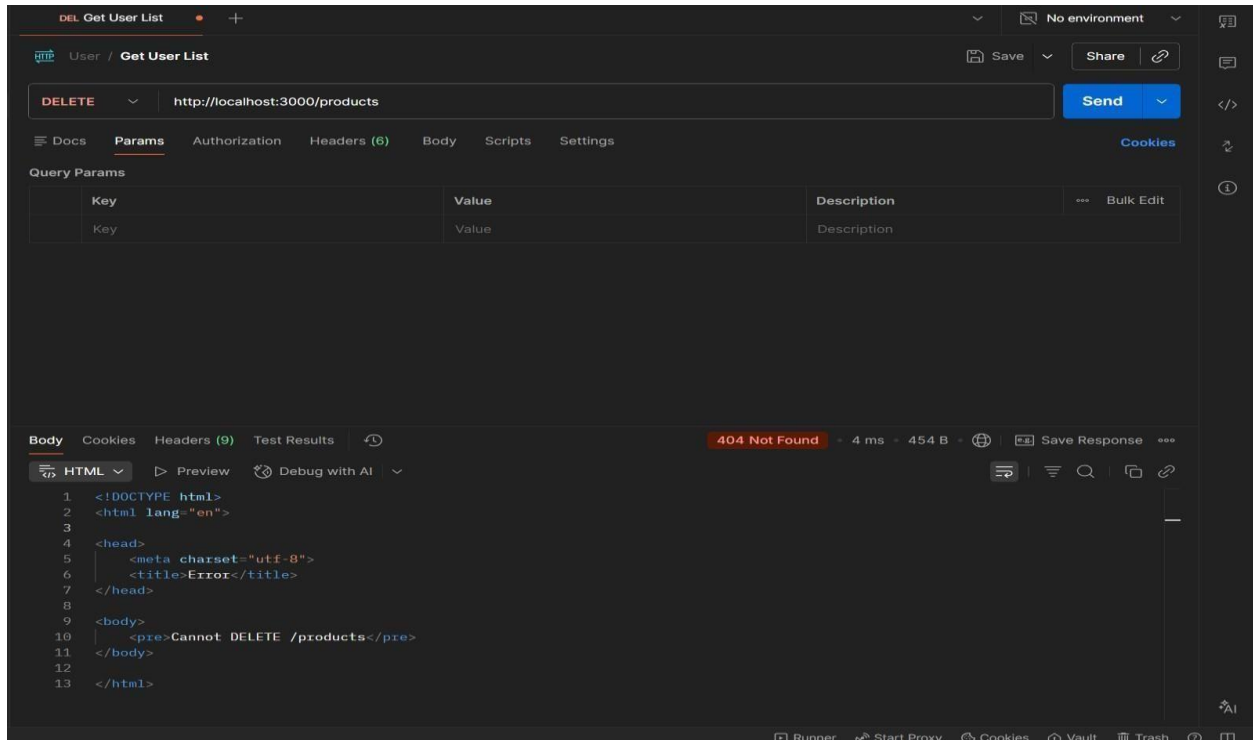
DB.JS:

```

let products = [];
function getAll() {
return products;
}
function add(p) { products.push(p);
}
function update(id, newData) { products
= products.map(p =>
p.id == id ? { ...p, ...newData } : p
);
}
function remove(id) { products =
products.filter(p => p.id != id);
}
module.exports = { getAll, add, update, remove };

```


OUTPUT:



RESULT: The RESTful API for Product Management was successfully implemented with GET, POST, PUT, and DELETE methods returning JSON responses.

TASK14: MICROSERVICES APP

AIM: To design a simple microservices-based system by splitting a monolithic application into independent services with loose coupling and single responsibility.

ALGORITHM:

1. Start the program
2. Identify modules in monolithic system
3. Split system into independent services
4. Create separate folders for each service
5. Implement Service 1 (Product Service)
6. Implement Service 2 (Order Service)

7. Use separate databases (db.js) for each service
8. Assign different ports for each service
9. Enable communication via REST APIs
10. Keep services loosely coupled
11. Deploy and run services independently
12. Access services and verify functionality

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Microservices App</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2>Microservices Demo</h2>

  <input id="pname" placeholder="Product Name">

  <button onclick="addProduct()">Add Product</button>

  <input id="order" placeholder="Order Item">

  <button onclick="addOrder()">Place Order</button>

</div>

<script src="script.js"></script>

</body>
```

</html> **STYLE.CSS:**

body { font-family: Arial;

background: #1e1e2f;

color: white;

display:flex; justify-

content:center; align-

items:center;

height:100vh;

}

.container { text-

align:center;

}

input { margin:

5px;

padding:

8px;

}

button { padding: 8px;

margin: 5px;

background:

#00c6ff; border:

none;

}

SCRIPT.JS:

function addProduct() {

 fetch(["http://localhost:3001/products"](http://localhost:3001/products), {

```

    method: "POST", headers: { "Content-Type":
    "application/json" }, body: JSON.stringify({ name:
    val("pname") })
  });
}
function addOrder() {
  fetch("http://localhost:3002/orders", { method:
  "POST", headers: { "Content-Type":
  "application/json" }, body: JSON.stringify({ item:
  val("order") })
  });
}
function val(id) { return
  document.getElementById(id).value;
}

```

SERVER.JS:

```

const express = require("express");
const cors = require("cors"); const
db = require("./db"); const app =
express(); app.use(cors());
app.use(express.json()); // GET
products app.get("/products",
(req, res) => { res.json(db.getAll());
});
// ADD product
app.post("/products", (req, res) => {

```

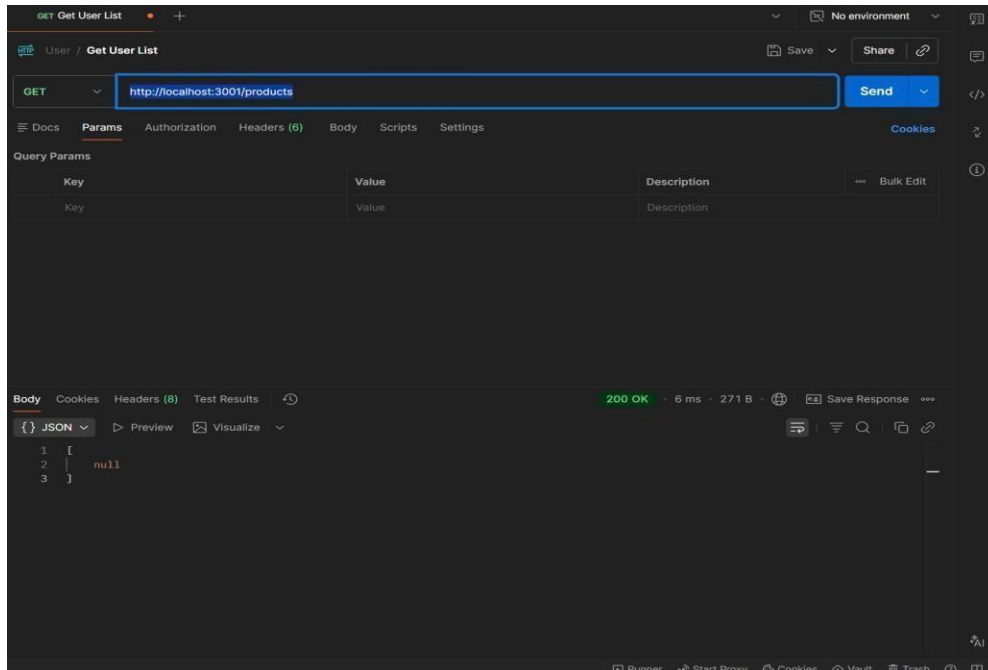
```
db.add(req.body); res.json({ msg:
"Product added" });
});
app.listen(3001, () => { console.log("Product Service running at
http://localhost:3001/");
});
```

DB.JS:

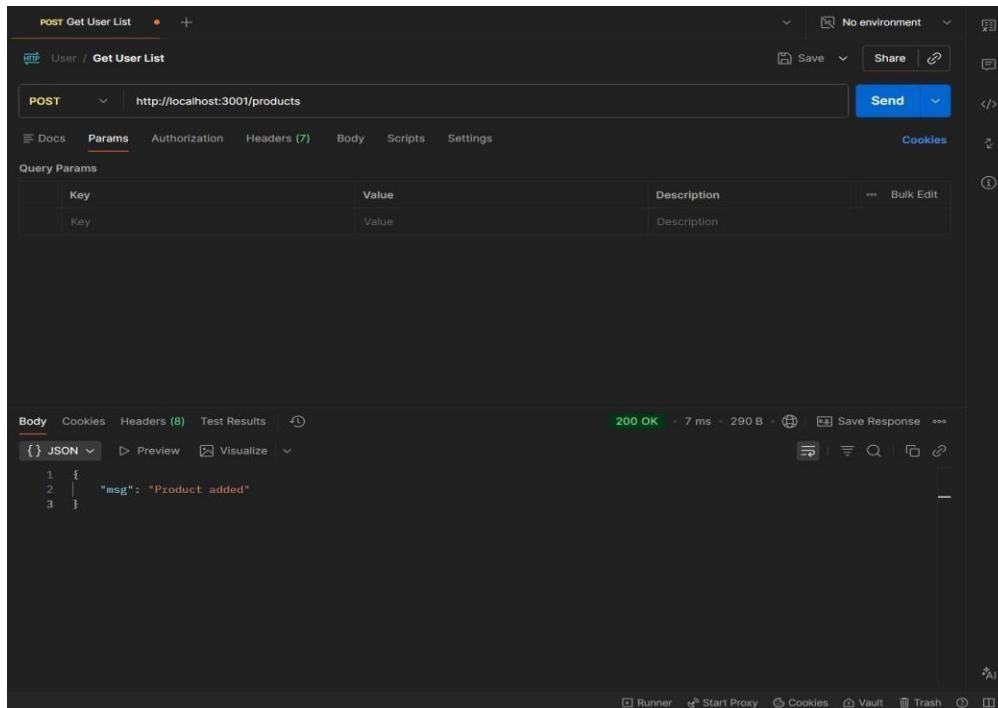
```
let products = [];
function add(p) {
products.push(p);
}
function getAll() { return
products;
}
module.exports = { add, getAll };
```

OUTPUT:

1. GET : <http://localhost:3001/products>



2. POST: <http://localhost:3001/products>



RESULTS: The application was successfully divided into independent microservices with separate responsibilities and communication via REST APIs

TASK15: Service Registry App

Aim: To implement service registry and discovery using Node.js by dynamically registering and discovering microservices without hard-coded URLs.

ALGORITHM:

1. Start the program
2. Create a registry server
3. Store service name and URL in memory
4. Start Product Service
5. Register Product Service to registry
6. Start Order Service
7. Register Order Service to registry
8. Maintain service list in registry
9. Client requests service from registry
10. Fetch service URL dynamically
11. Call the discovered service API
12. Stop the program

PROGRAM:

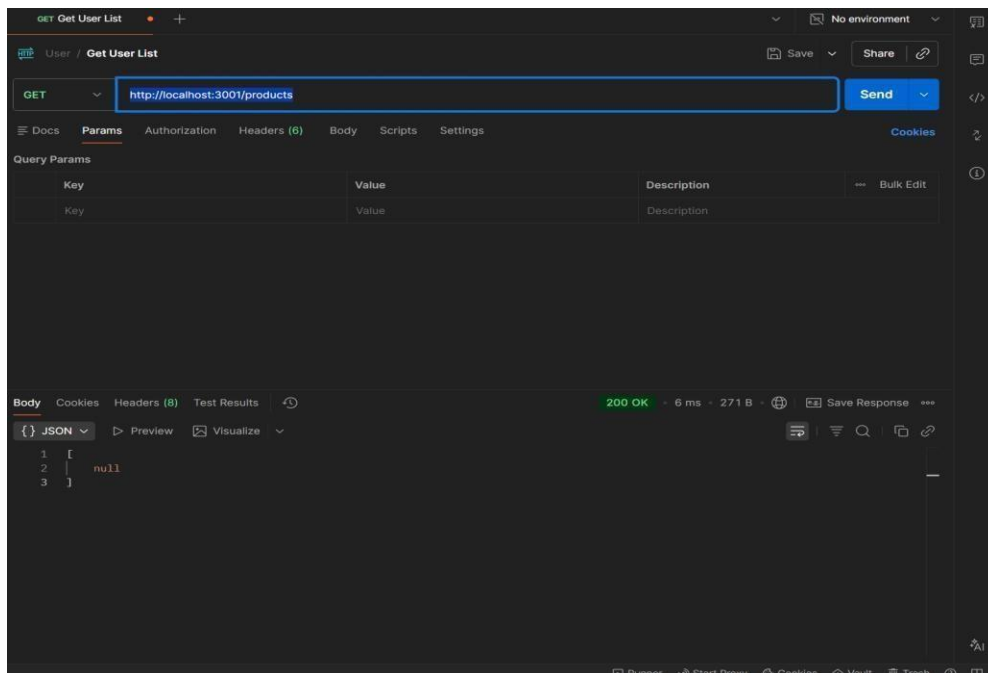
```
SERVER.JS:  const  express  =
require("express");  const  app  =
express();  app.use(express.json());
let      services      =      {};
app.post("/register", (req, res) => {
const { name, url } = req.body;
services[name]      =      url;
console.log(`${name} registered at
```

```

    ${url}`);    res.json({    msg:
    "Registered" });
});
app.get("/service/:name", (req, res) => { const
    service = services[req.params.name];
    if (!service) { return res.status(404).json({ error: "Service
    not found" }) res.json({ url: service });
});
app.listen(4000, () => { console.log("Registry running at
http://localhost:4000/");
});

```

OUTPUT



RESULT: The microservices-based system was successfully implemented with service registry and discovery, enabling dynamic communication between independent services without hardcoded URLs.

TASK16: API GATEWAY API

AIM: To configure an API Gateway that routes client requests to microservices and performs basic load balancing across multiple service instances.

ALGORITHM:

1. Start the program
2. Create API Gateway server
3. Define routes for microservices
4. Create multiple instances of a service
5. Store service URLs in an array
6. Implement round-robin load balancing
7. Receive client request at gateway
8. Select service instance dynamically
9. Forward request to selected instance
10. Get response from service
11. Send response back to client
12. Stop the program

PROGRAM:

```
SERVER.JS:  const express =  
require("express"); const axios =  
require("axios");  const app =  
express(); const services = [  
"http://localhost:3001",  
"http://localhost:3002"  
];
```

```

let index = 0; app.get("/products", async
(req, res) => {
  try { const service = services[index]; index = (index + 1) %
    services.length; const response = await
    axios.get(service + "/products");
    res.json(response.data);

  } catch (err) { res.status(500).json({ error: "Service
    unavailable" });
  }
});
app.listen(5000, () => { console.log("API Gateway →
http://localhost:5000/products");
});

```

OUTPUT:

```

{
  "instance": "Service 1",
  "data": ["Laptop", "Phone"]
}

```

```

{
  "instance": "Service 2",
  "data": ["Tablet", "Watch"]
}

```

RESULT: The API Gateway was successfully implemented to route requests and distribute load across multiple service instances using round-robin load balancing.

TASK17: Inter Service App

AIM: To enable inter-service communication where one microservice consumes another service's REST API and handles responses and failures gracefully.

ALGORITHM:

1. Start the program
2. Create Product Service (provider)
3. Create Order Service (consumer)
4. Implement REST API in Product Service
5. Store product data in memory
6. Implement Order Service API
7. Use axios/fetch to call Product Service
8. Receive response from Product Service
9. Process and store order data
10. Handle errors using try-catch
11. Send proper success/error response
12. Stop the program

FOLDER STRUCTURE:

```
inter-service-app/  
|  
├── product-service/  
|   └── server.js  
|  
├── order-service/  
|   └── server.js
```

PROGRAM:

PRODUCT-SERVICE:SERVER.JS:

```
const express = require("express");

const app = express(); let products

= [

  { id: 1, name: "Laptop" },

  { id: 2, name: "Phone" }

];

app.get("/products", (req, res) => { res.json(products);

});

app.listen(3001, () => { console.log("Product Service →

http://localhost:3001/products");

});
```

ORDER-SERVICE:SERVER.JS:

```
const express = require("express");

const axios = require("axios"); const

app = express();

app.use(express.json()); let orders = [];

app.post("/orders", async (req, res) => {

try {

  const response = await axios.get("http://localhost:3001/products");

  const products = response.data; const order = { id: Date.now(),

    products: products

  };

}
```

```

orders.push(order); res.json({ message:
"Order created successfully", order:
order

});
} catch (error) { res.status(500).json({ error: "Failed to
fetch products from Product Service"

});
}
});
app.get("/orders", (req, res) => { res.json(orders);
});
app.listen(3002, () => { console.log("Order Service →
http://localhost:3002/orders");
});

```

OUTPUT:

```

{
  "message": "Order created successfully",
  "order": {
    "id": 123456,
    "products": [
      { "id": 1, "name": "Laptop" },
      { "id": 2, "name": "Phone" }
    ]
  }
}

```

```
1  [  
2    {  
3      "id": 1,  
4      "name": "Laptop"  
5    },  
6    {  
7      "id": 2,  
8      "name": "Phone"  
9    }  
10 ]
```

RESULT: Inter-service communication was successfully implemented using REST APIs, with proper handling of responses and failures between microservices.

TASK18: Microservice Test

AIM: To write unit test cases for microservices to validate service logic, REST APIs, and data handling using a testing framework.

ALGORITHM:

1. Start the program
2. Install testing framework (Jest, Supertest)
3. Configure test script in package.json
4. Create test folder/files
5. Import service modules
6. Write test cases for service logic
7. Write test cases for REST APIs
8. Mock external dependencies if needed
9. Execute test cases using Jest
10. Verify expected outputs
11. Handle edge cases and errors
12. Stop the program

PROGRAM:

app.js:

```
const express = require("express");
const app = express();
app.use(express.json()); let
products = [];
// API app.get("/products", (req,
res) => { res.json(products);
});
```

```

app.post("/products", (req, res) => {
  products.push(req.body);
  res.json({ msg: "Product added" });
});

module.exports = app;

```

SERVER.JS:

```

const app = require("./app"); app.listen(3001,
() => {
  console.log("Product Service running");
});

```

```

Product.Test.js: const request = require("supertest"); const app
= require("../product-service/app"); describe("Product API
Tests", () => { test("GET /products should return empty array",
async () => { const res = await request(app).get("/products");
expect(res.statusCode).toBe(200);

expect(res.body).toEqual([]);

});

test("POST /products should add product", async () => { const
  res = await request(app)
    .post("/products")
    .send({      name:      "Laptop"      });

  expect(res.statusCode).toBe(200);

  expect(res.body.msg).toBe("Product
  added");

});

test("GET /products should return data", async () => {

```



```
const res = await request(app).get("/products");  
expect(res.body.length).toBeGreaterThan(0);  
});  
});
```

OUTPUT:

```
PASS tests/product.test.js  
✓ GET /products should return empty array  
✓ POST /products should add product  
✓ GET /products should return data
```

RESULT: Unit test cases were successfully implemented to validate microservice logic, REST APIs, and data handling, ensuring application reliability.

TASK19: Mini Git System

AIM: To design a simple Git-like web application that tracks file changes, stores versions, and displays commit history using HTML, CSS, JavaScript, and Node.js.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI for entering file content
4. Capture user input using JavaScript
5. Send data to server using fetch API
6. Store versions in database (array)
7. Assign version number to each commit
8. Save commit message and timestamp
9. Create API to fetch commit history
10. Display commit history in UI
11. Allow viewing previous versions
12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Mini Git System</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>
```

```
<div class="container">

  <h2>Mini Git System</h2>

  <textarea id="content" placeholder="Write your code..."></textarea>

  <input id="message" placeholder="Commit message">

  <button onclick="commit()">Commit</button>

  <h3>Commit History</h3>

  <ul id="history"></ul>

</div>

<script src="script.js"></script>
```

```
</body>
```

```
</html>    STYLE.CSS:
```

```
body { font-family: Arial;
background: #1e1e2f;
color: white;
display: flex; justify-
content: center; align-
items: center;

height: 100vh;
}
```

```
.container { width:
400px; text-align:
center;
}
```

```
textarea { width:
100%; height:
```

```

    80px; margin:
    5px;
}
input, button {
    margin: 5px;
    padding: 8px;
}
li
{
    background: #333;
    margin: 5px;
    padding: 5px;
}

```

SCRIPT.JS:

```

function commit() {
    const content =
    document.getElementById("content").value; const message =
    document.getElementById("message").value;
    fetch("/commit", { method: "POST", headers: {"Content-
    Type":"application/json"}, body: JSON.stringify({ content,
    message })
    }).then(loadHistory);
}
function loadHistory() {
    fetch("/history")
    .then(res => res.json())
    .then(data => {

```

```

    let list = document.getElementById("history");
    list.innerHTML = ""; data.forEach(c => { let li =
    document.createElement("li"); li.innerText =
    `v${c.version}:          ${c.message}`;
    list.appendChild(li);

    });

});

}

loadHistory();

```

SERVER.JS:

```

const express = require("express");
const db = require("./db"); const
app = express();
app.use(express.json());
app.use(express.static(_dirname));
app.post("/commit", (req, res) => {
db.save(req.body); res.json({ msg:
"Committed" });

});

app.get("/history", (req, res) => {
    res.json(db.getAll());

});

app.listen(3000, () => {
    console.log("http://localhost:3000");

});

```

DB.JS:

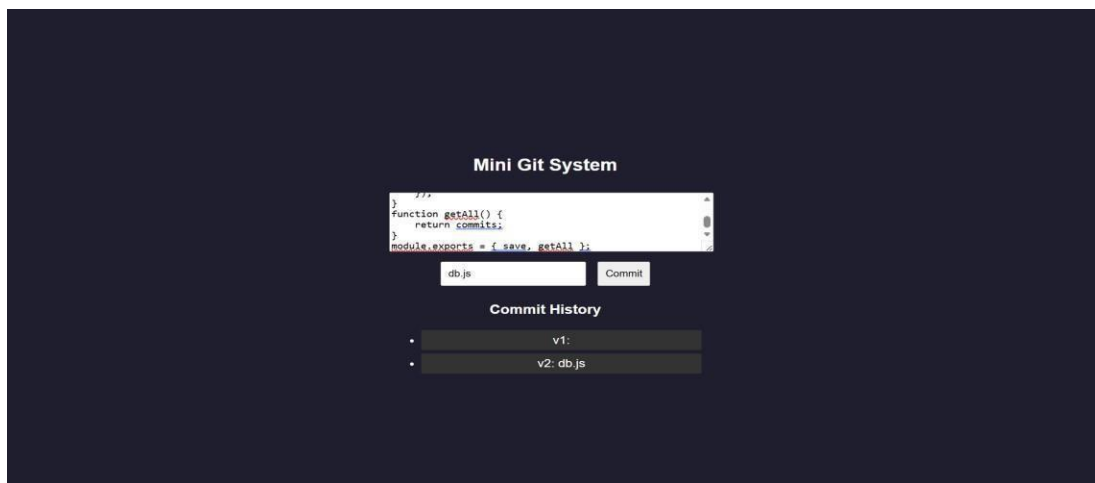
```

let commits = []; let version
= 1; function save(data) {
commits.push({    version:
version++,        message:
data.message,     content:
data.content,    time:  new
Date()

    });
}
function getAll() { return
    commits;
}
module.exports = { save, getAll };

```

OUTPUT:



RESULT: A mini Git-like system was successfully developed to track versions, store commits, and display history through a web interface.

TASK20: Mini Git Branch App

Aim: To develop a Git-like application that supports branching, merging, rebasing, and conflict resolution using a web interface.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Store branches and commits in memory
4. Create default branch (main)
5. Add feature to create new branch
6. Commit changes to selected branch
7. Implement merge logic
8. Detect conflicts (same file modified)
9. Show conflict message
10. Resolve conflicts manually
11. Implement rebase logic
12. Display branch history and stop

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Mini Git Branching</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">
```

```

<h2>Mini Git Branching</h2>
<input id="branchName" placeholder="New Branch Name">
<button onclick="createBranch()">Create Branch</button>
<input id="message" placeholder="Commit Message">
<button onclick="commit()">Commit</button>
<button onclick="merge()">Merge</button>
<button onclick="rebase()">Rebase</button>
<h3>Branches</h3>
<ul id="branches"></ul>
<h3>History</h3>
<ul id="history"></ul>
</div>
<script src="script.js"></script>
</body>
</html>

```

STYLE.CSS:

```

body { font-family: Arial;
background: #1e1e2f;
color: white;
display: flex; justify-
content: center; align-
items: center;
height: 100vh;
}
.container {
text-align: center;
}

```



```

input, button {
    margin: 5px;
    padding: 8px;
}

li
{
    background: #333;
    margin: 5px;
    padding: 5px;
}

```

SCRIPT.JS:

```

function createBranch() {
    const name =
        document.getElementById("branchName").value;

    fetch("/branch", { method: "POST", headers: {"Content-
        Type":"application/json"}, body: JSON.stringify({ name })
    }).then(load);
}

function commit() {
    const message =
        document.getElementById("message").value;

    fetch("/commit", { method: "POST", headers: {"Content-
        Type":"application/json"}, body: JSON.stringify({ message })
    }).then(load);
}

function merge() { fetch("/merge", {
    method: "POST" })

```

```

    .then(res => res.json())
    .then(alert)
    .then(load);
}
function rebase() { fetch("/rebase", {
    method: "POST" })
    .then(load);
}
function load() {
    fetch("/data")
    .then(res => res.json())
    .then(data => { let b =
        document.getElementById("branches"); let h =
        document.getElementById("history");
        b.innerHTML = "";
        h.innerHTML = "";
        data.branches.forEach(br => { let li =
        document.createElement("li");
        li.innerText = br;
        b.appendChild(li);
    });
    data.history.forEach(c => {
        let li = document.createElement("li"); li.innerText
        = c;
        h.appendChild(li);
    });
});

```

```
});  
}  
load();
```

SERVER.JS:

```
const express = require("express");  
const db = require("./db"); const app  
= express(); app.use(express.json());  
app.use(express.static(_dirname));  
app.post("/branch", (req, res) => {  
  db.createBranch(req.body.name);  
  res.json({ msg: "Branch created" });  
});  
app.post("/commit", (req, res) => {  
  db.commit(req.body.message);  
  res.json({ msg: "Committed" });  
});  
app.post("/merge", (req, res) => {  
  const result = db.merge();  
  res.json({ msg: result });  
});  
app.post("/rebase", (req, res) => { db.rebase(); res.json({ msg: "Rebased" });  
});  
app.get("/data", (req, res) => { res.json(db.getData());  
});  
app.listen(3000, () => console.log("http://localhost:3000"));
```

DB.JS:

```

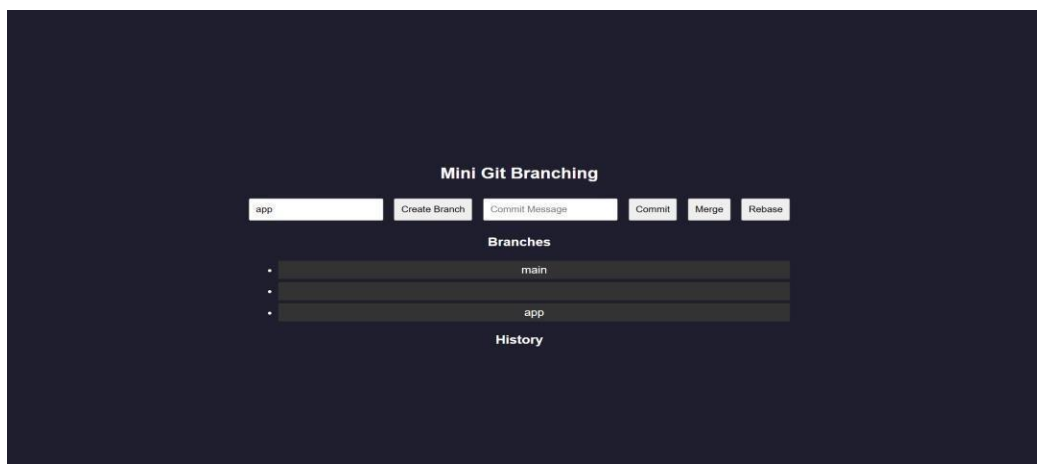
let    branches    =
["main"]; let current =
"main"; let commits = {
main: []
};
function    createBranch(name)    {
    branches.push(name); commits[name]
    = [...commits[current]];
}
function            commit(msg)            {
    commits[current].push(msg);
}
function    merge()    {    let    mainCommits    =
commits["main"];    let    featureCommits    =
commits[current];    if    (mainCommits.length    &&
featureCommits.length)    {    return    "Merge    Conflict
Detected!";
    }
commits["main"] = [...mainCommits, ...featureCommits]; return
    "Merged Successfully";
}
function rebase() { commits[current] = [...commits["main"],
    ...commits[current]];
}

```

```
function getData() { return {
  branches,      history:
  commits[current]
};
}

module.exports = { createBranch, commit, merge, rebase, getData };
```

OUTPUT:



RESULT: A Git-like branching application was successfully developed to simulate branching, merging, rebasing, and conflict resolution through a web interface.

TASK21: Mini Cid App

AIM: To develop a CI/CD pipeline simulation application that automates build, test, and deployment steps through a web interface.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI for pipeline trigger
4. Capture user action (Run Pipeline)

5. Send request to backend
6. Simulate code checkout step
7. Simulate build process
8. Simulate test execution
9. Simulate deployment step
10. Store pipeline logs
11. Display pipeline status in UI
12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Mini CI/CD Pipeline</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2>CI/CD Pipeline Simulator</h2>

  <button onclick="runPipeline()">Run Pipeline</button>

  <h3>Pipeline Logs</h3>

  <ul id="logs"></ul>

</div>

<script src="script.js"></script>

</body>
```

</html> **STYLE.CSS:**

body { font-family: Arial;

background: #1e1e2f;

color: white;

display:flex; justify-

content:center; align-

items:center;

height:100vh;

}

.container { text-

align:center;

width: 400px;

}

button { padding: 10px;

margin: 10px;

background:

#00c6ff; border:

none; color: white;

cursor: pointer;

}

li

{

background: #333;

margin: 5px;

padding: 5px;

}

SCRIPT.JS:

```
function runPipeline() { fetch("/run",
    { method: "POST" })
    .then(() => loadLogs());
}

function loadLogs() {
    fetch("/logs")
    .then(res => res.json())
    .then(data => { let list =
        document.getElementById("logs");
        list.innerHTML = ""; data.forEach(log => {
            let li = document.createElement("li");
            li.innerText = log; list.appendChild(li);
        });
    });
}

loadLogs();
```

SERVER.JS:

```
const express = require("express");
const db = require("./db"); const app =
express(); app.use(express.json());
app.use(express.static(_dirname));
app.post("/run", async (req, res) => {
    await db.runPipeline(); res.json({ msg:
    "Pipeline executed" });
});
```



```
app.get("/logs", (req, res) => { res.json(db.getLogs());
});
```

```
app.listen(3000, () => {
  console.log("http://localhost:3000");
});
```

DB.JS: let logs = []; async function

```
runPipeline() { logs = []; logs.push("
```

```
Code Checkout..."); await delay();
```

```
logs.push(" Build Started..."); await
```

```
delay();
```

```
  logs.push(" Running Tests..."); await
```

```
  delay(); logs.push(" Tests Passed");
```

```
  await delay(); logs.push("• Deploying
```

```
  Application..."); await delay();
```

```
  logs.push(" Deployment Successful!");
```

```
}
```

```
function getLogs() { return
```

```
  logs;
```

```
}
```

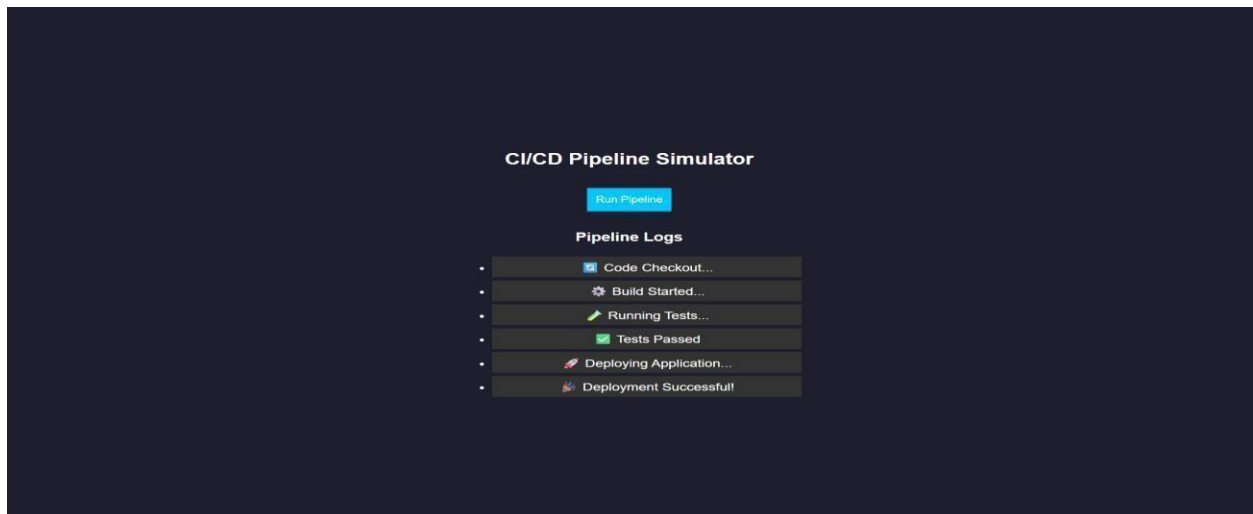
```
function delay() { return new Promise(resolve =>
```

```
  setTimeout(resolve, 1000));
```

```
}
```

```
module.exports = { runPipeline, getLogs };
```

OUTPUT:



RESULT: A CI/CD pipeline simulation application was successfully developed to automate and visualize build, test, and deployment processes.

TASK22: Cloud App

AIM: To develop a cloud comparison application that displays and compares services and pricing models of AWS, GCP, and Azure

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design user interface with dropdown for cloud providers
4. Initialize cloud service data (AWS, GCP, Azure) in database module
5. Capture selected provider using JavaScript
6. Send request to server to fetch service details
7. Display compute, storage, database, and networking services
8. Accept user input for compute hours and storage usage
9. Send input data to server for cost calculation
10. Calculate cost based on pay-as-you-use pricing model
11. Display total cost dynamically on UI

12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Cloud Comparison</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h2>Cloud Service Comparison</h2>

  <select id="provider" onchange="loadData()">

    <option value="aws">AWS</option>

    <option value="gcp">GCP</option>

    <option value="azure">Azure</option>

  </select>

  <h3>Services</h3>

  <ul id="services"></ul>

  <h3>Pricing Calculator</h3>

  <input id="hours" placeholder="Compute Hours">

  <input id="storage" placeholder="Storage (GB)">

  <button onclick="calculate()">Calculate Cost</button>

  <h3 id="result"></h3>

</div>

<script src="script.js"></script>
```

```
</body>
```

```
</html>      STYLE.CSS:
```

```
body { font-family: Arial;
```

```
background: #1e1e2f;
```

```
color: white;
```

```
display: flex; justify-
```

```
content: center; align-
```

```
items: center;
```

```
height: 100vh;
```

```
}
```

```
.container { text-
```

```
align: center;
```

```
width: 350px;
```

```
}
```

```
input, select, button
```

```
{ margin: 5px;
```

```
padding: 8px;
```

```
}
```

```
li
```

```
{
```

```
background: #333;
```

```
margin: 5px;
```

```
padding: 5px;
```

```
}
```

```
SCRIPT.JS:
```

```

function loadData() {
  const provider =
    document.getElementById("provider").value;
  fetch(`/services/${provider}`)
    .then(res => res.json())
    .then(data => {
      let list =
        document.getElementById("services");

      list.innerHTML = "";

      Object.entries(data).forEach(([key, value]) =>
        {
          let li = document.createElement("li");
          li.innerText = `${key}: ${value}`;
          list.appendChild(li);
        }
      );
    });
}

function calculate() {
  const provider =
    document.getElementById("provider").value;
  const hours =
    document.getElementById("hours").value;
  const storage =
    document.getElementById("storage").value;
  fetch("/cost", {
    method: "POST",
    headers: {"Content-Type": "application/json"},
    body: JSON.stringify({ provider,
      hours, storage })
  })
    .then(res => res.json())
    .then(data => {
      document.getElementById("result").innerText = "Cost:
        $" + data.cost;
    });
}

```

```

}

loadData();

SERVER.JS: const express = require("express");
const db = require("./db"); const app =
express(); app.use(express.json());
app.use(express.static( _dirname));

app.get("/services/:provider", (req, res) => {
res.json(db.getServices(req.params.provider));

});

app.post("/cost", (req, res) => { const { provider,
  hours, storage } = req.body; const cost =
  db.calculate(provider, hours, storage); res.json({
  cost });

});

app.listen(3000,
  ()
  =>
  {
    console.log("http://localhost:3000");
  }
});

```

DB.JS:

```

const services = { aws:
  {
    Compute: "EC2",
    Storage: "S3",
    Database: "RDS",
    Network: "VPC"
  },
  gcp: {

```

```

    Compute: "Compute Engine",
    Storage: "Cloud Storage",
    Database: "Cloud SQL",
    Network: "VPC"
  },
  azure: {
    Compute: "Virtual Machines",
    Storage: "Blob Storage",
    Database: "SQL Database",
    Network: "VNet"
  }
};

const pricing = { aws: { compute: 0.1,
  storage: 0.02 }, gcp: { compute: 0.09,
  storage: 0.025 }, azure: { compute:
  0.11, storage: 0.018 }
};

function getServices(provider) { return
  services[provider];
}

function calculate(provider, hours, storage) { const p =
  pricing[provider]; return (hours * p.compute + storage *
  p.storage).toFixed(2);
}

module.exports = { getServices, calculate };

```

OUTPUT:

Cloud Service Comparison

AWS ▾

Services

EC2

Pricing Calculator

9

8

Calculate Cost

Cost: \$1.06

Cloud Service Comparison

GCP

Services

•

Compute: Compute Engine

•

Storage: Cloud Storage

•

Database: Cloud SQL

•

Network: VPC

Pricing Calculator

9

8

Calculate Cost

Cost: \$1.01

Cloud Service Comparison

Azure

Services

•

Compute: Virtual Machines

•

Storage: Blob Storage

•

Database: SQL Database

•

Network: VNet

Pricing Calculator

9

8

Calculate Cost

Cost: \$1.13

RESULTS: A cloud comparison application was successfully developed to display services and simulate pay-as-you-use pricing across AWS, GCP, and Azure.

TASK23: AI PROMPT APP

AIM: To develop an application that demonstrates prompt engineering by generating and evaluating code or cloud configurations based on user input.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI with input field for user prompt
4. Capture user prompt using JavaScript
5. Send prompt to server using fetch API
6. Receive prompt data in backend (server.js)
7. Process prompt using predefined logic (db.js)
8. Generate output (code / cloud config / response)
9. Evaluate output quality and assign score
10. Send generated output and score to frontend
11. Display output and score on UI
12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>AI Prompt App</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">
```

```
<h2>Prompt Engineering Tool</h2>

<textarea id="prompt" placeholder="Enter your prompt..."></textarea>

<button onclick="generate()">Generate Output</button>

<h3>Generated Output</h3>

<pre id="output"></pre>

<h3>Quality Score</h3>

<p id="score"></p>

</div>

<script src="script.js"></script>

</body>

</html>    STYLE.CSS:

body { font-family: Arial;
background: #202038;
color: white;
display: flex; justify-
content: center; align-
items: center;

height: 100vh;

}

.container { width:
400px; text-align:
center;

}

textarea { width:
100%; height:
```

```

    100px;
    margin: 5px;
}
button { padding:
    10px; margin: 10px;
    background:
    #00c6ff; border:
    none; color: white;
}
pre { background:
    #333; padding:
    10px;
}

```

SCRIPT.JS:

```

function generate() { const prompt =
    document.getElementById("prompt").value;
    fetch("/generate", { method: "POST", headers: {"Content-
    Type":"application/json"}, body: JSON.stringify({ prompt })
    })
    .then(res => res.json())
    .then(data => {
    document.getElem
    entById("output").i
    nnerText =
    data.output;
    document.getElem

```

```

entById("score").in
nerText =
data.score + "/10";
});
}
SERVER.JS:  const  express  =
require("express");  const  db  =
require("./db");

const app = express();

//                               Middleware
app.use(express.json());
app.use(express.static(_dirname)); // serve HTML

// Home  route  (optional  but  useful)
app.get("/", (req, res) => { res.sendFile(
dirname + "/index.html");
});

// Generate output app.post("/generate", (req, res)
=>      {      const      result      =
db.processPrompt(req.body.prompt);
res.json(result);
});

```

```
// START SERVER WITH LINK
app.listen(3000, () => { console.log("•
Server running at:"); console.log("
http://localhost:3000");

});
```

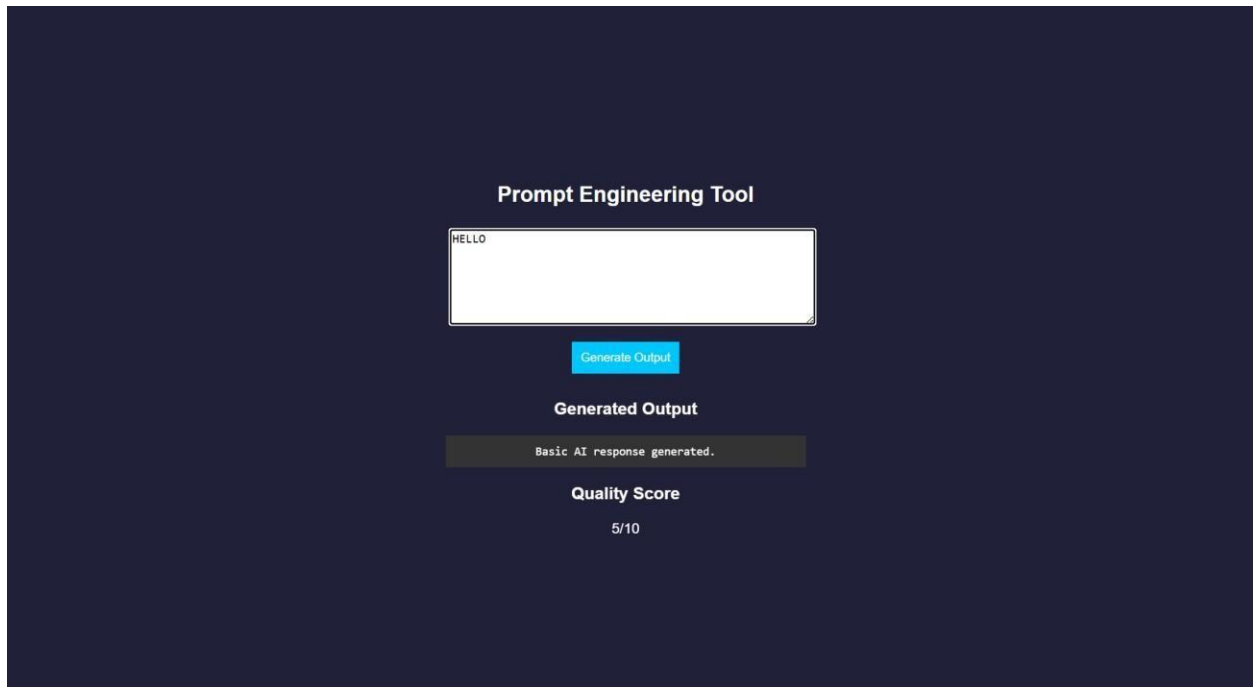
DB.JS:

```
function processPrompt(prompt) { let output =
  ""; let score = 5; if
  (prompt.toLowerCase().includes("code")) {
    output = `function greet() {
      console.log("Hello from AI!");
    }`;
    score = 9;
  }
  else if (prompt.toLowerCase().includes("cloud")) { output
    = `AWS EC2 Config:
- Instance: t2.micro
- Region: us-east-1`; score = 8;
  }
  else { output = "Basic AI response
    generated."; score = 5;
  }

  return { output, score };
}
```

```
module.exports = { processPrompt };
```

OUTPUT:



The screenshot displays a web application titled "Prompt Engineering Tool" on a dark blue background. It features a white text input field containing the word "HELLO". Below the input field is a blue button labeled "Generate Output". Underneath the button, the text "Generated Output" is displayed. A dark grey box contains the text "Basic AI response generated.". At the bottom, the text "Quality Score" is shown, followed by a score of "5/10".

RESULT: A prompt engineering application was successfully developed to generate and evaluate AI-based outputs for code and cloud configurations.

TASK24: Vibe Cloud App

AIM: To develop an application that uses iterative prompt engineering to generate and evaluate REST APIs, cloud deployment steps, and security configurations.

ALGORITHM:

1. Start the program
2. Create project files (HTML, CSS, JS, server.js, db.js)
3. Design UI to accept user prompts
4. Capture prompt input using JavaScript
5. Send prompt to backend using fetch API
6. Receive prompt in server and forward to processing module
7. Store prompt in history for version tracking
8. Analyze prompt type (REST API / Cloud / Security)
9. Generate corresponding output based on prompt
10. Evaluate output quality and assign score
11. Return output and score to frontend and display results
12. Stop the program

PROGRAM:

INDEX.HTML:

```
<!DOCTYPE html>

<html>

<head>

  <title>Vibe Coding App</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">
```



```

<h2>Vibe Coding Cloud Generator</h2>

<textarea id="prompt" placeholder="Enter prompt..."></textarea>

<button onclick="generate()">Generate</button>

<h3>Output</h3>

<pre id="output"></pre>

<h3>Score</h3>

<p id="score"></p>

<h3>Prompt Versions</h3>

<ul id="history"></ul>

</div>

<script src="script.js"></script>

</body>

</html>

STYLE.CSS:

body { font-family: Arial;
background: #b98bdb;
color: white;
display: flex; justify-
content: center; align-
items: center;

height: 100vh;

}

.container { width:
400px; text-align:
center;

}

```

```

textarea { width:
    100%; height:
    100px;
}
button { padding:
    10px; margin: 10px;
    background:
    #00c6ff; border:
    none;
}
pre { background:
    #333; padding:
    10px;
}
li
{
    background: #444;
    margin: 5px;
    padding: 5px;
}

```

SCRIPT.JS:

```

function generate() { const prompt =
    document.getElementById("prompt").value;
    fetch("/generate", { method: "POST", headers: {"Content-
    Type":"application/json"},

```

```

        body: JSON.stringify({ prompt })
    })

    .then(res => res.json())
    .then(data => { document.getElementById("output").innerText =
        data.output; document.getElementById("score").innerText =
        data.score + "/10"; loadHistory();

    });
}

function loadHistory() {
    fetch("/history")

    .then(res => res.json())
    .then(data => { let list =
        document.getElementById("history");

        list.innerHTML = ""; data.forEach((p, i) => { let
        li = document.createElement("li"); li.innerText
        = `v${i+1}: ${p.prompt}`; list.appendChild(li);

        });

    });
}

loadHistory();

SERVER.JS: const express =
require("express"); const db =
require("./db");

const app = express();
app.use(express.json());

```

```

app.use(express.static(_dirname));
app.post("/generate", (req, res) => { const
result = db.process(req.body.prompt);
res.json(result);

});
app.get("/history", (req, res) => {
  res.json(db.getHistory());
});
app.listen(3000, () => { console.log("•
http://localhost:3000/");
});

```

DB.JS:

```

let history = []; function process(prompt) { let output = ""; let
score = 5; history.push({ prompt }); if (prompt.includes("REST")) {
output = `app.get('/api', (req,res)=>res.send('API running'))`;
score = 7;

  }

  else if (prompt.includes("cloud")) { output
    = `Deploy using AWS EC2:

1. Launch instance

2. Install Node.js

3. Run app`; score = 8;

  }

  else if (prompt.includes("security")) { output = `Use HTTPS, JWT
    authentication, and firewall rules`; score = 9;

  }

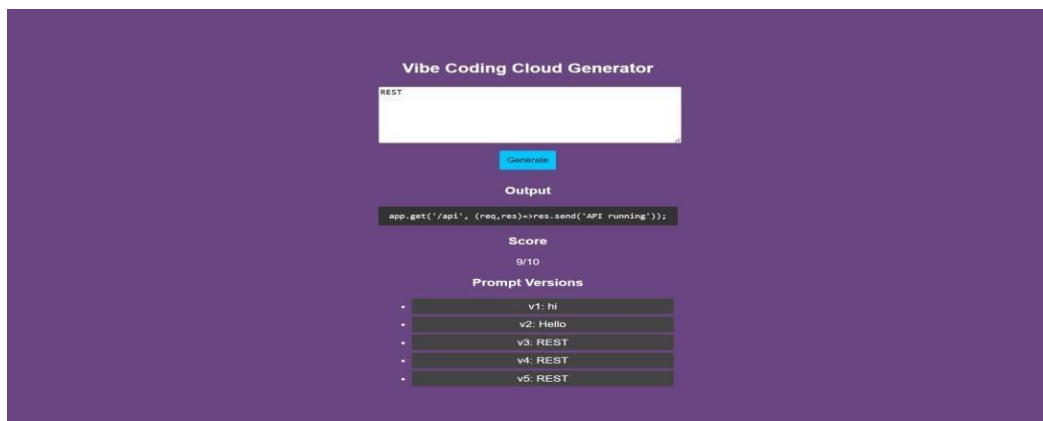
```

```

    score += Math.min(history.length, 2); return
    { output, score };
}
function getHistory() { return
    history;
}
module.exports = { process, getHistory };

```

OUTPUT:



RESULT: A Vibe Coding application was successfully developed to generate and refine cloudbased features using iterative prompt engineering.